

# PSY502 – History & Systems of Psychology

## HIGHLIGHTED HANDOUTS Lesson 1 – 22

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# History & Systems of Psychology PSY502

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**LESSON 01****INTRODUCTION****Course Objectives**

The objectives of this course are to introduce the students to the developmental history of the subject of psychology, to prepare students to appreciate and use more advanced materials of psychology and to provide the basic and the most modern knowledge related to psychology.

**Methodology**

Main features of a given topic shall be discussed supported by reference materials and examples from everyday life. You might be quizzed verbally by the teacher asking a question, letting you write the answer and then giving you the correct answer. For example, “Freud established the Behaviorist School?” Right or wrong? The correct answer is of course “wrong.” Watson’s name is associated with Behaviorist School. At the end of two/three lectures a homework assignment would be given. Homework assignment would be in the form of a short essay-type exam. It might also consist of multiple choice questions (MCQs). There would be a mid-term exam after the completion of 21, one hour lectures. There would be a final exam after the completion of 45 lectures.

**Grading**

Midterm exam will carry 35% weightage of marks for the total course. Final exam will be a comprehensive exam and will carry 50% weightage of the total course. A total of eight assignments shall be given carrying 15% weightage of the total marks in the course.

**Texts**

Two main text books will be used for the course. The first one will be:

- “A History of Modern Psychology” by Duane Schultz & Sydney Schultz and
- the Urdu text book used will be “Nafsiyat ka Irtika,” by Rafiq Jaffer and Humair Hashmi.

Both books are easily available in the market. Handouts for the entire course, prepared by the instructor will also be available for your benefit.

**Course Overview up to Midterm**

A review of the ancient philosophy / psychology in the Indo-Pak Sub-continent, we will then have a look at the ancient Greek philosophy/psychology and go on to read about philosophy/psychology in the 5<sup>th</sup> to 12<sup>th</sup> century AD. We will then consider the impact of renaissance on psychology; it will thoroughly reviewed; we then look at the British Associationism and go on to review the French and German contributions to psychology up to the 18<sup>th</sup> and 19<sup>th</sup> Century. A review of Reflexive Psychology will follow and then we will see the impact of physical sciences on psychology. We then look at the Structuralist School of psychology, followed by a discussion on the Functionalist School of psychology, we will then review psychology in the beginning of 20<sup>th</sup> century with the American contribution that is known as “Behaviorism,” and review “**Behaviorism**” and “**Neo-Behaviorism**.” Midterm exam will be held for course covered from “Ancient philosophy/psychology to Neo-Behaviorism”

### Course Overview up to Final Exam

After the mid term exam more recent European contributions to psychology shall be discussed those will include: The Gestalt School of psychology, the English Dynamic psychology, the Historico-Evolutionary Psychology. We will then take a scientific outlook at mental illness. And then study the Psychoanalytical School in the beginning of 20<sup>th</sup> century, followed by Jung's Analytical Psychology, and Adler's Individual Psychology, followed by Psychodynamic contributors. It will then lead to a review of Humanistic Psychology. We will then see the modern trends in psychological treatment of mental disorders, including Anti-psychiatry Movement. We shall then see the development of psychology in parts of Africa and Asia in 20<sup>th</sup> Century and finally a look at psychology in the 21<sup>st</sup> century.

### Ancient Philosophy/Psychology in the Sub-Continent

The first and the foremost problem in any discipline is that of the scope and the definition. Such is the case with psychology. It is only when we define psychology that we can move forward to view "psychological" thoughts and ideas of man from ancient times up to today.

Psychology for that and the present purpose can be defined as the "study of mind and body." The ancient man regarded "soul" as something that was invisible and resided inside a person. He had his own explanations for the behavior of the soul and the body. For example the soul escaped the body when a person was asleep. The escaped soul then performed certain actions on behalf of the person that was visible in the form of dream work. Further, dream work, fainting spell, epileptic fits, mental and physical discrepancies were also regarded as expressions of the soul.

The ancient man also believed in what is termed "animism." Animism is the process of seeing human qualities in inanimate objects such as moon, stars, stones etc. He humanized those objects in order to make them easy to understand. The ancient man had a strong belief in magic and tried to learn and adopt magical tricks which were his way of exercising control over things around him.

Ancient Indo-Pak history of psychology was no exception where also magic was an important element. Magic as mentioned earlier, was used to exercise control over others and nature. It may be used for negative purposes or positive purposes.

The ancient man also indulged in different exercises and activities to gain control over his body. Yoga was one of such exercises involving breath control and different body postures. One of the yoga's beliefs was that body and soul are related and connected and this connection influences body and soul both.

The ancient philosophy or psychology of the sub-continent advanced the belief that consciousness is related to the brain. The ancient Europeans, the Greeks for instance, thought that consciousness was a function of the heart of a man.

About a thousand years B.C. the sub-continentals differentiated between mental and physical labor. Mental labor was the process of thinking and using the mind to solve problems, while physical labor was to perform certain tasks physically. Mental labor was considered superior by the ancient man while physical labor was considered inferior. They also believed that our mental faculties are inherited.

According to the sub-continentals Atma or the soul of a person was the pure self. Further, the world around us was an illusion or a deception played on us by our senses. Some others believed that consciousness and mind were the products of material development. These two beliefs are called Idealism and Materialism today. The later ancient Greek philosophy has a number of similarities to the sub-continental views.

**LESSON 02****ANCIENT GREEK PHILOSOPHY/PSYCHOLOGY**

The ancient Greek philosophers/psychologists regarded three elements to be the basic ingredients of all matter including humans. These three elements were:

- Water
- Air
- And fire

The ancient Greek philosophers also put forward the view that contradiction is a permanent element in the world and because of it the world is in flux all the time. This means that every time a phenomenon arises there is a counter explanation to the phenomenon. Thus a contradiction in explanation arises which leads to further investigation and further phenomenon and counter phenomenon. When this process continues over a long period of time, we find the world to be in a constant flux.

One of the greatest developments of the Ancient Greek philosophers was that they put forward the view that everything consists of small indivisible particles and these particles act upon on the soul to create sensations. This was later found to be true with the discovery of atoms.

**Socrates**

Socrates is considered to be one of the most important ancient philosopher/psychologist. He laid the foundation of ideas for many philosophers/psychologists to follow. Socrates was born in 469BC and he died in 399BC. In his opinion knowledge and truth reside in the mind and one has to look for and find it there. It means that in order to gain knowledge and understand various phenomenon one has to look inwards instead of looking at the outward things and trying to find an explanation in them.

He further said that knowledge and awareness create virtue, so in order to be virtuous one has to look into the self. In other words, Socrates stressed on the study of soul i.e. looking inwards, and he rejected external observation.

Another of the Socratic contributions is that he defined the concept of happiness. To Socrates, stealing is not happiness but the satisfaction gained through acquiring by honest means is happiness.

He regarded the soul, called psyche in Greek, as superior to body and said that the psyche or the soul helps us to distinguish between good and bad. Socrates asserted that all things have a definite purpose and nothing occurs without that purpose.

Another of the great contributions of Socrates is his method of teaching. It is called the Socratic method of teaching. It is also the modern method of teaching and it consists of a dialogue, rather than monologue and focuses on the logical correctness of the argument. The teacher and the student interact with each other and have an intelligent dialogue that helps them to understand and comprehend better. The monologue is discouraged because it involves only one way flow of information, further it does not clear up any misunderstanding that may arise in the mind of the students, who are not able to express themselves. The Socratic method of teaching is in practice these days.

**Plato**

Plato was a pupil of Socrates and another important Greek philosopher/psychologist. He was born in 427BC and he died in 347BC.

Following the Socratic tradition, Plato was also of the view that knowledge is innate and inside the mind. In order to extract knowledge one needs to look into the mind rather than looking at external objects. This method was later called the **introspective method** in which outwardly observation is discouraged and looking into the mind or the soul is encouraged. This method was later adopted by many philosophers/psychologists who agreed with Platonian and Socratic methodology.

Plato is also considered an idealist philosopher who **believed in the supremacy of ideas**. He did not believe in acquiring knowledge by empiricism and observing facts. He thought that ideas are the only source and the true source of knowledge. As an individual looks into his ideas he can extract the best possible solutions and explanations to the problems that are encountered by him or her.

Another of Plato's contributions is that he described the parts of personality as

- Intellect
- Will
- And appetite.

The intellect is the ability to understand, while the **will** is the drive to do something. **Appetite** on the other hand is the part of the personality that deals with bodily needs. According to Plato, ideas are eternal, they are not born, and neither do they die while worldly objects change and die. Therefore, true knowledge, as mentioned earlier, can be gained from ideas rather than observation. Plato also regarded the soul or psyche to be permanent and the body as something that could change. This further suggested that knowledge could be acquired through the soul or the psyche but not by the use of bodily sensory organs. To Plato sensory organs were a hindrance to the acquisition of knowledge.

Plato suggested that the soul has three parts:

- Reason, located in the head
- Passion, located in the chest
- Appetite, located in the stomach

The function of the **reason** is to control and direct the passion and appetite. **Passion** is the desire to perform a certain action. **Appetite part** of the soul refers to the natural needs of the body, such as hunger, thirst etc. According to Plato, some passions and appetites are satisfied in dreams. This concept given by Plato is similar to what was later given by Freud as id, ego and super ego. Plato was the one who established the first ever university by the name of Academy.

**LESSON 03****GREEK THINKERS****Aristotle**

Aristotle was one of the greatest philosopher/thinker ever in the history of mankind. This Greek philosopher/psychologist, Aristotle was born in 384BC and he died in 322BC. Aristotle was a student of Plato and his ideas not only show the influence of Plato's thinking but also sometimes Aristotle has presented completely opposing views to his teacher.

Aristotle wrote a book by the name of De Anima which means "the Soul." Since soul was considered to be the primary topic of interest for philosophers/psychologists at that time, Aristotle is considered to be the first psychologists ever. Aristotle rejected the idea of dualism of soul and the body and put forward the view that these two are not separate but one entity.

Aristotle was of the view that ideas in the human beings tend to become associated with each other and there are three reasons for the association of ideas which he gave:

**Similarity**

Similarity means that the mind tends to see certain similar patterns between two things and relates them together. This is the simplest explanation of the association of ideas where, for example, two events which a person witnesses have certain common elements. These may then be associated with each other due to the similarities present in them.

**Contiguity**

Contiguity refers to two things or events happening close to each other so that the person is able to relate them to each other. For example, if two events occur consecutively, one after the other, they tend to get associated with each other. This is an example of contiguity of ideas, where one even may remind of the other event which is contiguous to it.

**Contrast**

Contrast means that two events or ideas are completely opposing to each other, which results in the formation of a link between the two. Humans tend to associate the ideas together. These may be two conflicting statements by someone. One statement would remind a person about the other conflicting statement immediately.

Therefore, in this way ideas tend to become associated with each other.

According to Aristotle, man is a biological being whose functions include:

- Eating
- Perceiving
- Having children
- Thinking
- And doing things.

All of these are characteristics of living organisms. It eats to fulfill the requirements for the nourishment of its body and its growth. It perceives and responds to stimuli according to its perception. It has the ability to procreate and expand its species. Higher form of living beings such as humans has the ability to use their mind to think and analyze objects and happenings around them. Further, according to Aristotle, one of the functions of human beings is doing things, which means that humans are involved in various activities which they perform throughout their lives which may be anything such as reading this text.

According to Aristotle's teacher Plato, ideas are eternal. They are neither born nor do they die while worldly objects change. Therefore, according to him the true source of knowledge is ideas themselves. Plato further asserted the soul or psyche to be permanent and the body as something that could change. Therefore, according to Plato, knowledge could be acquired through the soul or the psyche and sensory organs were a hindrance to the acquisition of knowledge. This was an introspective and subjective method of learning, which means to look into the mind to find solutions to problems. Aristotle on the other hand rejected his teacher's views on introspection and said that true knowledge can only be gained through observation and empiricism. As human beings observe the phenomenon going on around them and the world around them, they tend to make judgments and decisions that are based on their objective perception rather than their subjective biases. Therefore, instead of looking to the mind, one needs to look at the world around in order to gain knowledge. This implies that in Aristotle's views, sensory organs are not a hindrance to learning, but they are the source of gaining knowledge.

Aristotle also described two types of human motives which are

- Primary motives

Primary motives are basic motives which are the desires of all human beings and are mostly bodily desires such as hunger, sex, thirst, anger and need for rest. Such motives are not just possessed by human beings but may also be possessed by other living beings. An important aspect of such motives is that they are not learnt by the organisms; rather they are a part of the genetic makeup of all organisms. Humans are born with these motives, in other words these motives are inherited and they become the cause of propelling a person to perform a certain act or to do something. For example, hunger may cause a poor man to beg for food. Therefore, his basic bodily desire has caused him to perform a certain action. Such desires or motives are called primary motives by Aristotle.

- Secondary Motives

Secondary motives are those motives which are learnt by organisms and these become habits of the individuals. For example, it is the habit of certain people to socialize with others. Some people have the desire to dominate their fellows. These motives become the habits of individuals and they tend to follow them quite often. Therefore, they are called secondary motives by Aristotle.

Another great contribution of the great philosopher Aristotle was that he recognized that there are individual differences in various respects in people. These differences may be in the intelligence level, in certain abilities, sports, math etc. For example, one individual may be a very good sportsman but not a very good musician, while another may be a very good musician but not a sports man. Therefore, differences exist in all individuals as far as their personal characteristics are concerned. This is the reason for some individuals to excel in certain fields.

Aristotle further suggested that these individual differences are created during the life span of a person, but they are inherited by the person. He is born with these differences which may although be discovered in later in life.

Apart from these individual differences that are inherited and are a part of the genetic makeup of all individuals, Aristotle was of the view that the early childhood experiences and training of an individual have a profound impact on him or her. Childhood is the time when the mind is open to all influences from the environment. A child would learn to behave in a certain manner as he would see the individuals around him doing. Further, training of a child is also an important factor that would influence his behavior throughout his life. The way he has been trained and educated during the early years of his life, that is childhood, he is expected to show some influence of it in every part of his or her life. A child of Pakistani family who has been educated and grown up in the United States cannot be expected to behave in a similar manner to the one who has been grown up in Pakistan. That is because his experiences in his environment and training that he gets is absolutely different from the child grown in Pakistan. This influence is expected to last through out his or her life.

Aristotle also gave the concept of Catharsis. Catharsis is the sudden release of emotions by a person when he sees, hears or feels something. For example a person may start crying when he hears about a tragedy that someone else has encountered, or a people often have soaked eyes when they watch a tragic scene in a movie. These are the examples of catharsis where emotions are suddenly released by the individual. There may be also the emotions such as happiness or joy which someone feels when a villain is being beaten up by the hero of the movie. The list of contributions of Aristotle is a long one. This is why he is considered to be one of the greatest thinkers of mankind. His ideas have left a profound and a long lasting impact on man's thinking.

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**LESSON 04****PSYCHOLOGY IN THE 5TH TO 12TH CENTURY**

The period from 5th to 11th century A.D. is considered the dark ages of Europe. It was the period in between the great Greeks and the progressive industrialized Europe. The dark ages were the period when there was a general discouragement of research and investigation. Problems were not solved in the light of research and observation but by invoking religious edicts. Intellectuals adopted Plato's idealism while Aristotle's empiricism and observation were rejected.

It was believed that God was the most exalted being, followed by wisdom, followed by soul and then followed by matter. So matter of the body was the lowest form of existence and wisdom the highest after God. It was also thought that wisdom could be gained by looking inwards and by reflection which was the Socratic and Platonian point of view. The dark ages also had an impact on psychology. But during the same period emerged some prominent philosophers/psychologists whose contributions to psychology are noteworthy.

**Saint Augustine**

An important contribution to the subject of psychology of those times is that of Saint Augustine. He was born in 354AD and died in 430AD.

Saint Augustine proclaimed that evil or sin is a product of free will. This means that it is the free will of an individual that makes him indulge into sins, evils and crimes. There ought to be checks on every person's free will which is certainly what the order of the society is about. A society imposes certain limits to the exercise of free will of a person which therefore results in regulation of the society. If every person is allowed to practice his will freely there would be no society and only chaos. Saint Augustine further proclaims that pride is a cardinal sin which is also the cause of many other evil acts that a person indulges in. It makes him unaware of the consequences of his actions which may be harmful for others. Therefore, Saint Augustine points out certain psychological elements in human beings that influence their actions and the society as a whole.

Saint Augustine agreed with the prevailing ideas that the universe contains two basic principles:

- The principle of light and goodness
- And the principle of darkness and evil

This means that there are two paths in the universe and it is up to the individual to decide upon which path to take. According to Augustine we can discover these by looking inwards into our minds or what is called introspection. This idea is similar to that of the Socratic Platonian ideas. According to them, looking into the self brings awareness and light of knowledge and that is the only way of gaining knowledge. Plato even considered the sense organs as a hindrance to the acquisition of knowledge. Plato and Socrates had also said that knowledge is virtue. St. Augustine added to their views by saying that a person can look into self in the same way as one looks at external objects. How to introspect or to look into the self was one of the biggest difficulties that were encountered by the philosophers and thinkers of that time. St. Augustine provided a simple solution to it by saying that we can look at external objects through our sense organs, but we can look in by reflection. Reflection here refers to looking at and understanding ideas and thoughts and finding solutions to problems through the association of ideas within the mind. This was his solution to the problem of how to look inwards. Because of this stress by St. Augustine on looking inwards and his method of looking inwards, he is usually regarded as the first introspective psychologist. Introspective psychologists therefore are those psychologists who encourage looking into the mind to find solutions to the problems rather than observing outwardly things and trying to find solutions to problems by using the sensory organs, as propounded by Aristotle. Plato and Socrates on the other hand were in favor of introspection.

Another great contribution of St. Augustine is his explanation of consciousness which in his view is constantly flowing. This also is related to his idea of introspection. He said that consciousness flows constantly in the form of a stream and it is up to us how we look into the stream to get ideas and generate solutions to our problems. This stream of ideas is the basis of all knowledge. His ideas of introspection were later used by psychologists such as Wundt and Titchner.

### Avicenna

Avicenna holds very prominent position in the history of philosophy/psychology. He was born in 980AD and he died in 1037AD. Avicenna studied the Holy Quran, physics, theology and medicine and adopted a most rigorous method of the distinction between concepts.

Like the ancient thinkers, Avicenna was also of the view that human soul was non-material and immortal. It does not exist in any material form, but nevertheless exists in the body. Therefore he rejected the concept of a unity between soul and the body as given by Aristotle. To him the soul was a separate entity and the body was a separate one. Avicenna was of the view that soul has its identity and its own operative laws just as there are the operative laws of the body. In order to study the soul and discover solutions to problems regarding the soul and the body, we need to study and become aware of the operative laws of the soul. This in turn would then help in the process of gaining knowledge as well. Once the operative laws of the soul are determined, it would be easier to know how the soul thinks and what governs its actions and influences it to behave in a certain manner.

Avicenna was of the view that the mind has a tremendous effect upon the body, such that it can make it sick or make it healthy. Here the mind and the soul are considered to be the same thing as the ancient thinkers or philosophers thought. Avicenna thought that the soul has an influence to the extent that if a person thinks that he is not well the body responds in a similar fashion and he actually becomes ill. Therefore, soul is stronger than the body and it can dictate the body to behave in a certain manner. He further added that strong soul or mind can not only affect a person's own body but it can affect others as well. This is the case in hypnosis where a person tends to bring the other person under the influence of his mind and makes him do what he wants him to.

Elaborating his views on the strength of the mind or the soul, Avicenna said that the soul had the power to the extent that it ordered the body to move and the body obeyed. In other words the soul was responsible for all the actions of the body.

Following the Aristotelian tradition Avicenna recognized four stages of motion. Motion here refers to the process of performing a certain act. Based on these stages of motions, Avicenna tried to explain the human actions. The four stages of motion are:

#### Imagination

Imagination is the process of thinking about something in which in turn leads to a desire of possessing that thing.

#### Desires

It is the second stage of motion according to Avicenna which represents a person's willingness to achieve something or to act in a certain manner. The person desires to possess something or to do something.

#### Impulsion

The desires of the person cause him or push him to do something. This is the third stage of motion as given by Avicenna. The person is motivated by the desire to make a certain move which would satisfy the desire.

#### Movement

The last stage of motion according to Avicenna is the movement where a person actually makes the move to satisfy the desire that was caused by his imagination and which has forced him to perform a certain action or make a movement to satisfy it.

In other words, imagination creates desires, which in turn propels the person and then the person acts.

Another of Avicenna's contributions is that he distinguished between primary and secondary perceptions.

### **Primary Perception**

It is the subjective perception of a person that is based on his personal dispositions. For example if a person is taking a walk in a garden and he has in his mind that there are snakes in that garden, he is quite likely to confuse a twig with a snake. That is due to his personal disposition that there are snakes in that garden. In other words, an illusion is a primary perception. This kind of perception is called subjective perception and in Avicenna's terminology Primary perception.

### **Secondary Perception**

Secondary perception is objective perception based on logical reasoning and rationality of observation. In this case personal biases and disposition do not influence the perceptions which are totally based on empirical evidence.

Avicenna's idea of primary and secondary perception was similar to what was later proclaimed by the Gestalt School as "geographical reality" and "subjective reality."

As mentioned earlier, Avicenna showed the effect of soul or mind on the body. Therefore, he is considered to be a great healer and a physician. He also showed that some sick people recovered from illness by their will power and some fall sick because they lose the will to live. He also elaborated the concept "Wahm" which plays a significant part in Avicenna's thought. "Wahm" is akin to what modern psychologists, particularly psychotherapists describe as "nervous response." The modern psychotherapists have explained that nervous response is the forerunner of many psychological and physical ailments. This is what Avicenna had explained earlier. An example of nervous response or "Wahm" is the stress created in a person, which ultimately has negative impacts on a person's health, behavior and psyche.

Avicenna was the first one to develop a system which looked at diseases from not just the physical but also the psychological point of view. This is why he is considered to be a great physician as well as a psychologist.

**LESSON 05****PSYCHOLOGY IN THE 5<sup>TH</sup> TO 12<sup>TH</sup> CENTURY****Al-Ghazali**

Al-Ghazali was a great philosopher/psychologist. His ideas hold a unique position in the development of philosophy/psychology. Al-Ghazali was born in 1058AD and died in 1111AD. He studied Greek philosophy gaining mastery over it, and then studied religion and acquired a vast knowledge on that as well. Having a thorough knowledge of the subjects led Al-Ghazali to see a conflict between religion and philosophy. This conflict could only be seen by someone who had a vast knowledge of both the fields such as Al-Ghazali. The conclusion he drew, created so much stress and affected him so badly that he became physically ill. He continued his study for more than ten years and after much reflection on the subject, he was able to resolve the conflict for himself. He then gradually started to recover from his bad health.

Al-Ghazali had a unique idea about the soul of human beings. He taught that the soul was like God and said that “Allah has breathed His own spirit into man.” This entailed the soul being a higher expression of man’s existence. Therefore, Al-Ghazali also differed from Aristotle on the concept of unity of body and soul.

Another contribution of Al-Ghazali is that he distinguished between positive and negative virtues. In his views:

- Positive virtue is doing a good deed. For example if someone helps the poor that would be a positive virtue.
- And negative virtue is to avoid a bad deed. For example person avoids indulging into a criminal act.

Related to his idea of the soul being the higher expression of man’s existence, Al-Ghazali also distinguished between three stages of being which are:

- Lowest, the material world of existence
- The second is the physical world
- The third is the world of divine.

Al-Ghazali also gave a long list of vices such as hunger, sex, vice of speech, self assertion, love of wealth and hypocrisy. The vices of hunger and sex are physical needs. The vice of speech is to use bad words or to say such words which would harm someone. Another vice described by Al-Ghazali was a unique idea of self assertion which means that being over assertive and trying to dominate others and not listening to what they have to say. That is over asserting one’s own beliefs and thoughts. The next is the vice of love of wealth. According to Al-Ghazali like all other vices it makes a person forget about all moralities and values and he pursue his desires blindly. The last vice described by Al-Ghazali is the vice of hypocrisy which means that a person appears differently from what he actually is.

Al-Ghazali also described four kinds of love:

- Self love
- Love of a benefactor
- Love of beauty
- Love of harmonious interaction.

Self love is the love that an individual has for himself while the love for a benefactor is the love that an individual has for some one who takes care of his needs. For example the love for one's parents is the love for a benefactor. Third is the love for beauty which is possessed by every individual. Man has innate sense to love and admire beauty. This has been identified by Al-Ghazali. Last is the love for harmonious interaction which is the love for association with people. It is another of man's instincts which has been identified by Al-Ghazali.

One of Al-Ghazali's greatest contributions is that he proposed two kinds of selves:

- Nafsay-Ammarah; the primitive self or the negative, bad self
- And Nafsay-Mutminah; the satisfied self or positive good self.

Nafsay-Ammarah is original state of self in which man is dissatisfied. Nafsay-Mutminah is the satisfied self when the individual has achieved his potentials and desires and gotten rid of the bad self.

According to Al-Ghazali, man can change and improve himself from Nafsay-Ammarah to Nafsay-Mutminah by doing the right things which according to him are six:

- Masharaqa  
This means to put oneself straight. When the person decides that he would change himself and do those things only which are right or which are deemed right by the society.
- Maraqaba  
This means to guard oneself from evil and bad things.
- Mahasaba  
This means to criticize oneself for anything which is not good, which would ultimately lead to the improvement of that habit or doing.
- Maaqaba  
This means to punish the self for the wrong doing and forcefully restraining the self from it.
- Mujahida  
This means to express the self in fully and properly and not keep anything hidden.
- Maataba  
This means to cleanse oneself of any bad deed or doing.

In view of these ways of self discipline Ghazali is considered by many as a psychotherapist.

### **Ibn-Rushd, Averroes**

Averroes is regarded as one of the greatest philosopher/psychologist of the Middle Ages. He is also known as Ibn-Rushd in Urdu. He was born in 1126AD and he died in 1198AD. Unlike some of the previous philosopher/psychologists, he followed the Aristotelian tradition of observation and empiricism and based his ideas upon logic, far removed from speculation and Socratic and Platonic way of gaining knowledge by insight.

Averroes's views were different from that of Al-Ghazali and he held the view that soul is not spiritual but material and mortal. He further stated that soul does not survive at death. Therefore he rejected and ancient view of the soul being eternal and the body being mortal. That is what Socrates used as a basis for his method of gaining knowledge by insight.

Averroes divided soul into five categories:

1. **Nutritive**

The nutritive part of the soul is responsible for the growth and development.

2. **Sensitive**

The sensitive part of the soul is responsible for emotions and feelings.

3. **Imaginative**

The imaginative part of the soul is responsible for the thinking process of the individual.

4. **Cognitive**

This part is responsible for association of ideas and for the process of solving problems.

5. **Appetitive**

This part of the soul is responsible for bodily needs such as hunger thirst etc.

Averroes taught that animals gain knowledge by sense organs, whereas man does it by intellect and knowledge gathered by intellect is the true and correct knowledge. Therefore his method of gaining knowledge was a combination of the Socratic and the Aristotelian tradition where Socrates thought that sensory organs were a hindrance to the acquisition of knowledge while Aristotle was of the view that knowledge needs to be gathered through empiricism rather than looking inwards.

Averroes further asserted that when a change occurs in the world, knowledge also changes. Therefore, knowledge is in a constant flux. Hence, man needs to constantly update his knowledge.

Averroes was of the opinion that all events have causes and denial of causes is denial of knowledge. This is known as determinism. He concluded that knowledge could therefore be gained by looking at causes of objects and events. In light of this theory, even accidents which apparently do not have a cause actually have a cause. For example a car colliding into a tree has apparently no cause, but there has to be a cause for the accident in the view of Averroes. The cause therefore may be that the car was being driven at a high speed or the steering wheel was turned in the wrong direction.

Averroes practiced the Aristotelian way of observation and rejected speculation as a way of acquiring knowledge avoiding unnecessary speculation even in religious matters. Later Europeans adopted Averroes's ideas and rejected speculation which was the reason for their civilization to advance at a rapid pace.

**LESSON 06****RENAISSANCE**

Renaissance is a French word meaning “rebirth.” The term Renaissance is used to describe the development of western civilization that marked the transition from medieval to modern times. It is the period between 13<sup>th</sup> to 16<sup>th</sup> centuries AD. Earlier Europe had passed through what was called the dark ages the period between 5<sup>th</sup> and 11<sup>th</sup> centuries A.D., when all researches and fresh ideas were discouraged. Problems were not being solved in the light of research and observation but by invoking religious edicts. Intellectuals adopted Plato’s idealism while Aristotle’s empiricism and observation were rejected.

During Renaissance new ideas and fresh research was encouraged. There was a movement to search for ancient texts that would increase current scientific knowledge. Botany, zoology, magic, alchemy, astrology, and psychology were developed during the Renaissance as a result of the study of ancient texts. Scientific thinkers such as Leonardo Vinci and Galileo attempted to refine earlier findings. Many other great names also belong to this period, such as Francis Bacon, Newton and Kepler. As the Dark ages had an impact on the growth of philosophy/psychology so did the Renaissance.

**Rene Descartes**

An important name in this galaxy is that of a French philosopher/psychologist, Rene Descartes. He was born in 1596 AD and he died in 1650 AD. He was the shining star of the Renaissance; a soldier, scientist, mathematician, philosopher and a teacher. Descartes' methodology was a major influence in the transition from medieval science and philosophy to the modern era. In science, Descartes discarded tradition and supported rationalization and logic rather than ideas based upon experiences. Mathematics was his greatest interest; building upon the work of others, he originated the Cartesian coordinates and Cartesian curves; he is often said to be the founder of analytical geometry. He wrote a text on physiology, and he also worked in psychology; he contended that emotion was finally physiological at base and argued that the control of the physical expression of emotion would control the emotions themselves. His chief work on psychology appeared in 1649.

Descartes’ main method of acquisition of knowledge was to doubt everything. This meant that to him the existence of everything was doubtful and anything that he is feeling or he is seeing may be an illusion. By logically, following his skepticism and his doubt Descartes had come to the conclusion that he could doubt everything. For example, when a person sits on a chair, according to Descartes, the chair may not actually exist and it might just be a deception of his or her senses. If a person is eating something, the food might not be there and it might just be his or her senses which are making him believe that he is eating. Following his skepticism and his theory of doubting everything, Descartes reached the conclusion that he could doubt everything in this world except for one; that was the fact that he is doubting. In other words doubting according to Descartes was beyond doubt. This conclusion further led him to conclude that a person can doubt everything in this world except for his thinking.

Based on these conclusions, Rene Descartes went on to say that because he thinks therefore he exists. This conclusion was beyond doubts, a correct conclusion. This means that the existence of an individual is indicated by his capacity and ability to think. His thinking distinguished him from other individuals. Descartes gave his conclusion as a French term: “Cogito ergo sum.” It means that “I think therefore I am.” In other words, once again thinking signifies the existence of a person.

In the light of this argument, he concluded that thinking or ideas are above everything else. In other words the thoughts are the exalted entities while the actions of the person below them.

Descartes went on to divide the human being into two parts based on his conclusions of superiority of thoughts. The two parts of the human beings are:

- The part that thinks
- The part that acts

The thinking part was obviously the mind or the soul. It thinks, contemplates, and makes decisions for the body. Then these decisions are conveyed to the part that has to act, which obeys the orders of the thinking part or the mind. The part that acts is the body of the human being. This meant, based on the conclusion that thinking was above anything else, that the mind or the soul was superior to the body. Descartes further proclaimed that the mind or the soul were eternal while the body was mortal. This concept was similar to that given by Plato and Aristotle that the mind or soul was a permanent entity.

Another great contribution of Rene Descartes was, that based on his conclusion about mind and the body, he proclaimed that the body is like a machine that obeys the orders of the mind. We can discover the operative laws of the body as we can discover the operations of a machine. This meant that Descartes supported the separation of body from the soul. The ancient Greek philosopher Aristotle had put forward the view of the unity of the soul and the body, but Descartes refuted his claim. This separation of body from the soul is called "Dualism" in psychology.

Further, Descartes was of the opinion that the soul is metaphysical while the body is physical. The soul of the person and his mind interact and influence each other. This is also proven by the fact that the soul thinks and orders the body which obeys, while on the other hand the body feels and receives stimuli for the soul. In his view, the soul affects the body by a gland that he called "Pineal Gland," which he thought was the seat of the mind.

Descartes also distinguished between two types of thoughts:

- Innate thoughts
- And acquired thoughts

Innate thoughts are thoughts that we are born with. In other words these thoughts are inherited. On the other hand some thoughts are acquired thoughts that a person acquires through the course of his or her life, based on the experiences that he encounters.

Descartes is considered to be the first European who put forward the mechanical view of the body, which was that the body is like a machine which obeys the orders of the mind. He also proclaimed that the operative laws of the body were like that of a machine and could be discovered.

**LESSON 07****ASSOCIATIONISTS**

Associationist is a group of three English philosophers/psychologists of the 16<sup>th</sup> century. Associationists are given this name because they believed in the concept of association of ideas. They are very important in the history of psychology and their theories are known as **Associationism**.

**Thomas Hobbes**

One of the most important names in the context of Associationism is that of Thomas Hobbes. He was born on 5<sup>th</sup> April 1588 AD and died on 4<sup>th</sup> December 1679 in Wiltshire, England. Hobbes was a philosopher/psychologist and a writer who is known for his book "Leviathan" which was published in 1651. Hobbes is also known for his emphasis on materialism. Hobbes was educated at Oxford and worked as a tutor to the son of William Cavendish, later the Earl of Devonshire. His connections to the royal family gave him opportunities to travel and pursue his studies, but they also put him in the middle of the English Civil War. In 1640 political turmoil forced him to leave England for France, where he continued to associate with scholars and scientists of Europe, including Galileo and René Descartes.

As mentioned earlier, Hobbes is considered an important materialist thinker of the time. Materialism means, greater interest in material possessions rather than spiritual values. Materialist thinkers were of the view that matter is dynamic, which means that matter changes from one form to another. Hobbes put forward the view that all we see including humans are different forms of matter and the laws that govern matter are also applicable to animals and humans, therefore these laws can be discovered. He contented that animals and man are in fact different types of machines and their functions and operations can be studied, understood, explained and even predicted. This mechanical view of human beings was initially given by Rene Descartes. Hobbes went on to elaborate the idea given by Descartes by saying that all human feelings and emotions such as laughing, crying, feeling hungry could be understood and explained in terms of functions of the human machines. This means that there has to be an input into the human machine in order to get an output in the form of the above mentioned functions of the human machine. For example, when a person is subject to heavy stress, he may start crying. A person who has not eaten for a long time starts feeling hungry. Therefore, the human machine has to have some sort of an input in order to give the above mentioned responses as its functions or outputs. Hobbes concentrated on finding similarities between humans and machines and this idea of finding similarities between man and machines is called "mechanical materialism," and Hobbes was one of its originators.

Hobbes was of the opinion that our thoughts and emotions are caused by the motion the external stimuli. As the variables of the external stimuli change, they impact the human body which results in emotions and thoughts.

He further proclaimed that motion in the brain creates thoughts. Then these thoughts get accumulated and get connected. This results in creating chains of thoughts or ideas leading to complex ideas. This is what has been termed as the association of ideas. Aristotle has also given his explanation of association of ideas which according to him is caused due to similarity, contiguity and contrast. Hobbes and two other associationists propounded this concept later that ideas tend to get associated.

Thomas Hobbes believed that knowledge can be acquired by observation through the sense organs. Therefore, he rejected the idea of introspection and agreed with Aristotle's empiricism as a means of acquiring knowledge.

According to Hobbes three things create association:

- Habits

- Wishes
- Repetition

**Habits** result in association of ideas. **Wishes** create linkages in ideas. A person who wishes for something links many ideas with his wish which in turn results in association of ideas. Further, **repetition** of event also results in association of ideas. This is similar to Aristotle's contiguity and similarity when two things happening close to each other or having similarity with each other result in association of ideas.

Because of his views of association of ideas he is called an Associationist by historians of psychology.

### John Locke

Another important British Associationist was John Locke. He was born in 1632 AD and died in 1704 AD. Educated at Christ Church College, Oxford, he became (1660) a lecturer there in Greek, rhetoric, and philosophy. He studied medicine, and his acquaintance with scientific practice had a strong influence upon his philosophical thought and method. He is considered to be the founder of British Empiricism.

His major contribution in psychology came in the form of an Essay. In the Essay Concerning Human Understanding Locke examines the nature of the human mind and the process by which it knows the world. Repudiating the traditional doctrine of innate ideas, Locke believed that the mind is born blank and the world writes on it.

John Locke was of the view that ideas were not innate or inborn, but ideas were developed over the due course of time through experiences that an individual has. In other words he rejected the Platonian point of views of introspection and ideas being inborn. According to John Locke, the mind of a child at the time of his birth is like a clean slate. Locke called this clean slate a "tabula rasa." This means that the mind is open to all external influences. The slate is blank and anything could be written upon the slate. Experiences throughout the life of the individual write on the slate. This is how ideas and thoughts are formed. Therefore, ideas are not inborn; it is the environment around the individual that shapes his or her imagination, feelings and thoughts.

According to Locke, we learn by two ways:

- Sensations
- And reflection

**Sensation** is the acknowledgement of the stimuli by the body. For example if a person touches something hot, the body feels the hotness of the object and as a result person learns to avoid the hot object. This is learning through sensations, where a stimulus or may be stimuli induce the learning process to take place. On the other hand, **reflection** means the interpretation of the stimuli by the body. The individual interprets the stimuli according to his own perception. Therefore two persons may differ in their **reflection** i.e. the interpretation of the stimuli.

In Locke's views, sensations create ideas, but reasoning connects those ideas together and these connections or associations are the source of knowledge. When the sensations are interpreted, a reason for it is sought. This reason results in creating knowledge for the person. For example, the sensation of feeling something as hot may be interpreted by the person as the hot object being dangerous. Therefore, the person may reach the conclusion that hot objects are dangerous, the reason being that they hurt the body.

John Locke was also a political philosopher. He therefore contented that for the society to move forward, prosper and develop, it is essential to gather knowledge. Knowledge can be acquired through the process of sensation, reasoning and association of ideas. John Locke supported the Aristotelian view point of learning by empiricism rather than introspection. His method encouraged learning by observation. Because of his support for the theory of association of ideas, he is also considered as an associationist.

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**LESSON 08****ASSOCIATIONISTS****David Hume**

The third British Associationist was David Hume who was born in 1711 AD and died in 1776 AD. He was a philosopher and a historian. Educated at Edinburgh, Scotland he lived in France, where he finished his first philosophical work, *A Treatise of Human Nature*.

David Hume followed the Lockean viewpoint in saying that senses and observation are the true sources of knowledge and knowledge cannot be gained by looking into the mind. On the other hand Hume denied the existence of material world. The world is what the senses perceive it to be. For example, Hume was of the view that when a person sees a table lying in a room, the table exists because his sense of vision can detect it, but if the person goes away from the room the table does not exist anymore for him. Therefore, the existence of a material world is just a perception of the senses.

According to Hume, human mind tends to establish a cause and effect relationship. This means that if one event follows another, the first event is considered to be the cause and the second event is the effect. In other words, the apparent sequence of events in the external world is in fact the sequence of perceptions in the mind. From this statement Hume argued that our expectation that the future will be like the past has no basis in reason; it is purely a matter of belief. For example when we expect the sun to rise every morning, it is a belief based on the past experience.

Hume believed that the purpose of knowledge is to facilitate life. This is known as the Utilitarian approach according to which knowledge must serve life and it must have a purpose. Later James adopted a similar approach in purporting the idea of Pragmatism which means that everything has to have a practical application and reason.

Hume also distinguished between violent passions and calm passions. Violent passions for example are love and hate while calm passions are aesthetic and moral feelings.

**FRENCH REVOLUTION**

One of the greatest revolutions in the history of mankind is the French Revolution that took place in France in 1789. Throughout the 18<sup>th</sup> century, resentment increased in France against the nobles and the clergy, who lived lives of idleness and luxury. They paid no taxes at all while the peasants had to pay taxes to the church, to their local lord as well as the government. By the late 1780s the cost of the wars against England had left France bankrupt but except for the peasants still no one paid taxes. The result was a violent revolt against the government marking the end of monarchy and feudalism in France. It was one of the most important events in the history of mankind because it changed the outlook of man. It set an example for Europe and the rest of the world to follow. It had its impact on all the future developments of mankind, and ultimately the development of all disciplines including philosophy/psychology.

**Denis Diderot**

One of the important philosopher/psychologist of the era was Denis Diderot. He was French, born in 1713 AD and died in 1784 AD. Following the prevailing materialist point of view proposing that matter is dynamic and the movement of particles creates sensations and feelings in a body. It was similar to the viewpoint given by Thomas Hobbes. He postulated that thought and memory are the functions of the brain and man and animals possess memory. He was a follower of the Aristotelian tradition, believing that knowledge is gathered by observation.

Diderot said that knowledge and power come from observation and experimentation. He further proclaimed that the purpose of knowledge was not just to know the truth but it was to increase the power of man. Knowledge gave man the ability to understand things better and develop the power to judge things. It is only through observation and experimentation that man's thoughts and his conditions can be changed. Innate or inborn thoughts could not change the living conditions of man. By experimenting man can discover new techniques to make his life easy and comfortable. Therefore, his conditions could become better. This was later proved in the industrial revolution when research and development resulted in discoveries such as the steam engine which changed the history of man.

### Auguste Comte

Another important thinker/philosopher/psychologist of the period was Auguste Comte, a French philosopher. He was born in 1789 AD, died in 1875 AD. He was educated in Paris and is considered as the founder of the school of philosophy known as positivism. Comte is also considered as the first Sociologist. He made the first contributions to what we now call the subject of Sociology laying its foundations. Sociology is the study of large groups of people. He sees the intellectual development of man covered by what is called the Law of the Three Stages—first, in which events were largely attributed to supernatural forces; second, in which natural phenomena are thought to result from ideas; and third, in which phenomena are explained by observation, hypotheses, and experimentation.

One of the important ideas of Comte was that, that there is no difference in the chemical elements of trees, animals and man. This implied that the laws that govern trees and animals are also applicable to man; animals and trees need certain elements from the nature to survive. Similarly man also needs certain essential elements from the nature. In other words, man is also an organism that may be controlled and manipulated like trees or animals.

Unlike David Hume who gave the idea of cause and effect relationship, Comte taught that we should not be too concerned about cause and effect relationships. He purported that we should gather information and then try to formulate connections between different pieces of information or facts gathered. He also said that scientists should avoid moral or ideological judgments which make their study biased. They should focus and concentrate only on gathering facts and collecting information. Collection of information is the only method by which we can understand, control and predict events.

Comte proposed that man's mind passed through three stages of growth and development:

- First stage

This was when man explained and understood events by referring to the super natural, gods and spirits. This meant that some supernatural forces or powers make things happen. This was the most primitive stage of development. As the bases of phenomenon were not found, a general explanation of attributing them to the supernatural was used.

- Second stage

This stage was when man explained events in mechanical terms. It was when man realized that nature and man have their own mechanical laws, which are responsible for events. The discovery of the laws being the basis of man and natural phenomenon was the stage when man inclined towards empiricism and gaining of knowledge by observation and through the senses.

- Third stage

The third stage was the recent stage. It was when man understood and explained events in a deterministic way. Determinism is to explain events in terms of a cause and effect relationship. This meant that when two events occur, one after the other, the first event would be the cause of the second event. This cause and effect relationship was also given by David Hume.

Comte was primarily a social reformer. His goal was a society in which individuals and nations could live in harmony and comfort. His system for achieving such a society is presented in his book, “the Course of Positive Philosophy.” In this work Comte analyzes the relation of social evolution and the stages of science. It is in this book that he gave the three stages of development of man’s mind.

The sciences themselves are classified on the basis of increasing complexity and decreasing generality of application in the ascending order: mathematics, astronomy, physics, chemistry, biology, and sociology. Each science depends at least in part on the science preceding it; hence all contribute to “sociology” a term that Comte himself originated. These are the major contributions of Auguste Comte.

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**LESSON 09****GERMAN CONTRIBUTION****Wilhelm Leibniz**

The first German philosopher/psychologist in the present context was Wilhelm Leibniz. He was born in 1646AD and he died in 1716 AD.

One of the most important contributions of Wilhelm Leibniz is that the universe including human beings is made up of atoms. His theory was based on the preexisting belief of atoms forming the matter. Leibniz called these particles or atom as **Monads**.

Leibniz further proposed that Monads are spiritual in nature and they are permanent and indestructible. They have their own stages of growth and development. Leibniz said that objects are made up of lowest monads, animals of higher monads than objects and man is made of the highest monads of them all. We are born with these, and they develop with the passage of time.

Another contribution of Wilhelm Leibniz is that he proposed that the soul and the body exist and operate parallel to each other. This means that the body and the soul operate in the humans at the same time and complement each other in their functions. This view of parallelism of soul and body is termed as **psycho-physical parallelism**.

**Immanuel Kant**

Another great German philosopher/psychologist was Immanuel Kant who was born in 1724 AD and died in 1804 AD.

Kant was against introspection and believed that observation and empiricism can lead to knowledge. The important contribution of Kant in this regard is that he proposed that the human mind is active and it rearranges and catalogues information gathered by observation. This means that whenever an individual observes something, he stores the information in his mind. The mind performs the function of the storage house and also arranges the information for the individual. This may result in easy retrieval of the information and the information being associated with other pieces of information that are in the same category. Therefore, this combination of information results in formulation of ideas etc.

Kant proposed that the mind has an innate capacity to see unity in what is observed. This means that mind relates every observation and information to some other information and this is what was termed as unity. It finds different observations as similar because of the previously stored information.

Further, Kant proposed that mind has the capacity to give meaning to sensations and observation. The sensations received by the body are conveyed to the mind. Then the mind is responsible for interpreting these sensations and deciding how the person and the body have to react. Similarly for observation, the mind acquires all the information and tries to draw conclusions out of it. Therefore, the mind performs the function of an interpreter.

Kant's ideas are a mixture of Leibnizian and Lockean views. Leibniz proposed that in the sense both mind and body operate at the same time. Leibniz had proposed the concept of psycho-physical parallelism. On the other hand Kant's views were similar to Locke in terms that according to Kant there was not material existence of the mind and its operations. Kant further proposed that psychology could not become a science, since science deals in matter and mind is not material.

**Friedrich Hegel**

Another name belonging to the German epoch was that of Friedrich Hegel who was born in 1770AD and died in 1831AD.

Hegel taught that there is no conflict between mind or wisdom and nature or matter. He further proclaimed that there is a logical flow in nature, which can be discovered by logic and wisdom. This means that all the phenomenon of nature can be described in logical terms. It may have a cause and effect relationship or may otherwise be explained using simple methods of observational reasoning.

He postulated the view of Absolute idea. **Absolute idea** is the absolute truth and it is non-material. Hegel put forward the idea that objects and events can not be studied separately. This idea had the implication in his theory that if we want to study a person we must look into his family life, his political and religious ideas, his economic conditions etc. It is one of the most important features of Hegel's viewpoint. Hegel correctly and implicitly stated that the behavior of a person, his attitude and his actions are influenced by his environment. The kind of environment from which a person belongs to has to determine his own behavior. For example an individual living in a joint family system would have different attitude from one living in a separate family. He also purported that economic, political and social environment also influence the person, which can be understood by looking at the French revolution. The French peasants felt that they were being oppressed by the clergy men and the rulers. This made them revolt against both the church and the monarch.

Hegel proposed that objects and events affect each other and in turn get affected. Similarly human thought affects objects, and objects affect human thoughts. This means that humans tend to use their thoughts for the improvement or to make best use of the objects. On the other hand natural phenomenon related to the objects influence human minds and make humans develop new ideas and thoughts. In other words this conflict between objects and human thought leads to change, growth and development.

Hegel also proposed a theory of development of man's thought which states that development takes place in three stages. This theory is also based on the idea of the conflict between objects and thoughts. The three stages of development of thought are:

- The first stage is when man discovers some phenomenon. He tries to explain the phenomenon according to his own information. In other words, man discovers, generates an idea and calls it a thesis. This thesis is the basis of the thought or idea.
- Later when the idea has been proposed in the form of a thesis, further observation and interpretation results in the formation of its opposite or contradictory idea. This opposite or contradictory idea is called the anti-thesis
- Finally the two conflicting or contradictory ideas are merged together to form a new idea which is known as synthesis. This generates the explanation of the phenomenon.

In other words, there is first an explanation of the phenomenon. This explanation is then negated by another explanation. When both these explanations are combined it results in a final explanation which is called a synthesis.

As a whole these three steps by the development of a synthesis lead to growth and development. The synthesis ultimately also becomes a thesis and is then negated in the same manner as the original thesis. This process of negation of the synthesis as a thesis is called the negation of negation. It continues forever. Therefore, according to Hegel contradiction is basic mover of the entire universe, including man and ideas.

Old ideas generate new ideas, those are negated and new ideas are formed. This means that there are no permanent ideas in this world. Hegel also taught that we should look at the universe and see the logical connections between objects and ideas. These logical connections would explain various phenomena. This process is called the dialectical process. The dialectical process means that by looking at two things, i.e. the object and the ideas we can draw conclusions about them. These were the contribution of Friedrich Hegel.

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**LESSON 10****INDUSTRIAL REVOLUTION**

Perhaps the most important development in the last one thousand years was the Industrial Revolution of the late 18<sup>th</sup> and 19<sup>th</sup> centuries. This was the real beginning of our modern world. Its main features were:

- Hand to machine manufacture
- Home to factory production
- Natural power to mechanical power

It was also characterized by a movement of people from villages to new industrial cities and the rapid development of transport.

The steam engine was the heart of industrial revolution. The first really practical engine was invented by the Scotsman James Watt in about 1775. For the first time in history, people did not have to rely on human or animal muscle, wind or water, for power. Very quickly the steam engine was being used for many things other than powering textile machines. It was used to pump water from coal mines; to lift trucks of coal to the surface; to blow air into furnaces for making of iron; to grind clay for pottery, and to power new factories of all kinds. For over a hundred years, steam was the king of the industry.

**Steam Engine being used for transportation**

The rate, at which new inventions were made, accelerated. The industrial Revolution consisted, in some respects, of a chain reaction. The construction of machines required more iron; more iron required more coal; more coal required faster and more efficient transport; more transport required engineering skills of all kinds. The manufacture of larger quantities of textiles, for example, needed more chemicals; new chemicals needed more scientists; scientist needed more specialized education.

Large numbers of workers had to move to new towns from countryside: they needed transport, housing, water, sanitation, shops, entertainment, a police force and general organization. The increased amounts of trading led to the need for more services such as banking, the post and later transport systems.

The industrial revolution changed England, and then spread to France, then affected Germany, then Russia and America, then Japan and then the rest of the world. Machines replaced much of the manual labor, industry developed and flourished, means of communication became rapid, electricity was invented and used for commercial purposes and an industrial class emerged which replaced feudalism. New ideas and curiosity were encouraged, and man's outlook and his attitude changed.

The industrial revolution does a great deal to the modern world today. The effect of this revolution could be seen immediately on the whole outlook of man, and it ultimately affected all the future developments of mankind. Man's outlook became more empirical and more open towards new ideas and innovations.

### RUSSIAN CONTRIBUTIONS

#### Ivan Sechenov



Ivan Sechenov

The first among the Russian contributors was Ivan Sechenov. He was born on 1<sup>st</sup> August, 1829 and he died on 2<sup>nd</sup> November, 1905. He taught at Moscow University. Ivan was physiologist, named by Ivan Pavlov as "The Father of Russian physiology." Sechenov authored major classic "Reflexes of the Brain" introducing electrophysiology and neurophysiology into laboratories and teaching of medicine. Sechenov's major interest was neurophysiology (the structure of the brain). He showed that brain activity is linked to electric currents and was the first to introduce electrophysiology. His work laid the foundations for the study of reflexes, animal and human behavior, and neuroscience.

Sechenov explained psychological functions in terms of physiological responses. This means that all psychological responses of the brain have some physiological reason or basis. In other words, when the brain acts in a certain manner, it is in response to certain physiological changes that have taken place.

Sechenov said that the mind and soul are the functions of the brain and that there is no duality between soul and the body. Therefore, he rejected the prevailing view of soul and the body being separate and accepted the Aristotelian point of view of unity of body and soul.

Sechenov concluded that all psychological responses are the result of sensory stimulation and that all psychological responses manifest themselves in behavioral forms. In other words, whenever there is a

change in the environment, it is received by the sensory organs of the body which convey it to the brain. The brain then reacts to these messages. This reaction of the brain is manifested in the form of the behavior of the person. For example when a person touches a hot surface, the sensory organs in his or her skin transmit the information to the brain which in turn sends back the signal to the body to withdraw away from the hot surface. Therefore, the behavior of the person is the response to the physiological change.

Sechenov also put forward the view that there are certain parts in the brain that augment and facilitate certain responses, and are certain other parts which inhibit certain responses. We now know this to be true. Sechenov's ideas deeply influenced the ideas of psychology and formed the basis of many new researches in the subject.

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## RUSSIAN CONTRIBUTIONS

### Ivan Pavlov

Ivan Pavlov is one of the greatest psychologists of 20<sup>th</sup> century. His contributions to psychology formed the basis of many later discoveries in the field. Pavlov was born in Ryazan, Russia in 1849 and he died in 1936. His father was a man of church therefore Pavlov started off by studying religion. He later went on to study medicine. Initially he worked as a research associate at Leipzig, Germany, where later the first psychological laboratory was established. At a very young age, after working at Leipzig, Pavlov was appointed as a professor at the Imperial Institute of Experimental Medicine.

One of the greatest achievements of Ivan Pavlov was his research on digestive glands of human beings. When published, this research was regarded very highly in the academic circles. Ivan Pavlov has the honor of being the first and only psychologist who has ever received a Noble Prize. He was awarded the Noble Prize in Physiology and medicine in 1904 for his work on the digestive system. As the Soviet Revolution took place in 1917, the chances of any further research work being done by Pavlov became meek, as it was expected that the new government would not allow Pavlov to continue with the research. On the contrary, the new Soviet Government supported Pavlov. The government not only provided funds to Pavlov for his research, but also provided any necessary facilities that were needed to carry out the research.

One of the major contributions of Ivan Pavlov is that he began to see the linkage of physiology and psychology. This means that the physiology of a person, that is the way a person is made up, or the general structure of a person and his mind are linked up. They both influence each other. This finding of his was published in his book in 1923. This book was also highly regarded in the academic circles. This was the book in which Pavlov presented his ground breaking theory, and his most important contribution to the field of psychology; the concept of learning by conditioning. It has been elaborately explained in his book. It is actually the explanation of the link between psychology and physiology that Pavlov had discovered. Earlier Pavlov had studied the ideas put forwarded by Sechenov, another Russian psychologist, and was greatly impressed with them.

Pavlov conducted a series of experiments on animals and humans. His experiments of dogs resulted in the discovery of learning by conditioning. His concept of learning by conditioning was explained by him using the following basic terms:

#### a) Reflex

Reflex may be defined as an unlearned or an automatic response of the organism to a stimulus. For example, when a puff of air strikes the eye, the eye blinks; when a person touches something hot, he immediately withdraws his hand etc. In these examples, the puff of air and the hot object become the stimulus while the response is the blinking of the eye or withdrawing of the body. Importantly all these actions are involuntary and a natural reaction of the body towards stimuli. This reaction is what has been termed by Pavlov as reflex.

#### b) Unconditioned stimulus (UCS)

It is a stimulus that elicits a natural response. In other words, it is a stimulus which causes the body to respond to itself, although the body is not conditioned to it. For example, when a bell rings, a dog which is able to hear the sound of the bell pricks his ears in response to the bell. Therefore, the dog responds to an unconditioned stimulus. The dog has not learnt to respond to the stimulus in that manner.

#### c) Unconditioned response (UR)

It is a natural response of an organism to an unconditioned stimulus. For example, when a dog looks at food, its mouth starts to salivate. This means the stimulus that the dog has received from the

environment, it is not conditioned, but the dog reacts in a certain manner to the unconditioned stimulus.

d) Conditioned stimulus (CS)

According to Pavlov, when we combine a neutral stimulus with an unconditioned stimulus, it evokes a different response. This response has been termed as Conditioned Response which is a result of the conditioned stimulus. The conditioned stimulus is when a dog is presented with the ringing of a bell along with food. This is expected to cause the effect of salivation in the dog's mouth. In other words, the stimulus of food has been conditioned using the stimulus of the ringing of bell.

e) Conditioned response (CR)

As the conditioned stimulus is given to the organism, it reacts in a certain manner, which is known as the **conditioned response of the organism**. In the previous example, a dog was presented with the stimulus of the ringing of a bell and that of food. This resulted in salivation in the mouth of the dog. But later, if the dog is presented with the stimulus of the bell alone, it is expected to salivate. This means that the response of the dog has been associated with the conditioned stimulus rather than the original stimulus of the food. The dog would give the same response as it gave to the stimulus of the food.

Based upon these basic elements, Pavlov put forward the idea of how animals and human beings learn. According to him, when an unconditioned stimulus and a conditioned stimulus are repeatedly presented together and they invoke an unconditioned response, it would later result in, that only the presentation of a conditioned stimulus evokes the same response as before.

In simple terms, considering the example given earlier, if ringing of a bell and food are presented together to a dog, and these lead to salivation, after repeated trials when the bell alone is rung, it evokes the response of salivation because the dog has learnt to expect food with the ringing of bell. This is how Pavlov showed by his experiments that animals and humans learn. Based upon his experiments he also formulated some principles of learning which are as follows:

a) Acquisition

According to Pavlov, learning in humans and animals takes place when the paired stimuli results in a response from the organism and the response is reinforced. That is if a dog is presented with food and ringing of bell at the same time, it results in learning of a certain response by the dog.

b) Extinction

If the conditioned response is not reinforced, the response disappears. This process is known as **extinction of the conditioned response**. For example if a dog is given food with the ringing of bell and later the dog is not given food with the ringing of bell, the dog would stop expecting food with it. This would result in disappearing of the conditioned response of salivation by the dog.

c) Spontaneous recovery

When the conditioned response of an organism disappears because the conditioned stimulus is not paired with the unconditioned stimulus, **the process is extinction**. The organism would not show the conditioned response as long as the conditioned stimulus is not reinforced, but once reinforcement is introduced, the extinguished conditioned response is quickly recovered. This means that when a dog is not given food with the ringing of bell, it stops expecting food with it. But as soon as the ringing of the bell is accompanied with the food, the dog again starts expecting food with the bell. This is known as spontaneous recovery.

d) Generalization

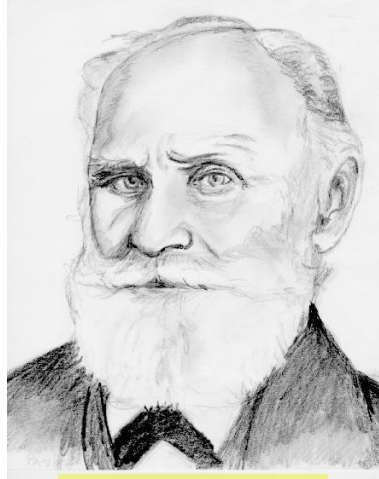
According to Pavlov, when similar stimuli evoke a conditioned response, it is a **case of generalization**. This means the dog which shows the conditioned response of salivating on the ringing of bell, when presented with a similar stimulus of ringing; it may show the same conditioned response as before.

e) Discrimination

When a slightly different stimulus does not evoke a conditioned response it is a case of discrimination. This means that for example, the dog is given the stimulus of the ringing of a bell with food using a special type of bell. When it has learned a conditioned response, the type of bell is changed and a similar bell is used. This then does not result in the conditioned response (salivating) of the dog. In other words, the dog has discriminated between the two types of stimuli.

This was the theory of learning presented by Pavlov. Some other contributions of Pavlov are also of significance in the field of psychology which shall be explained later.

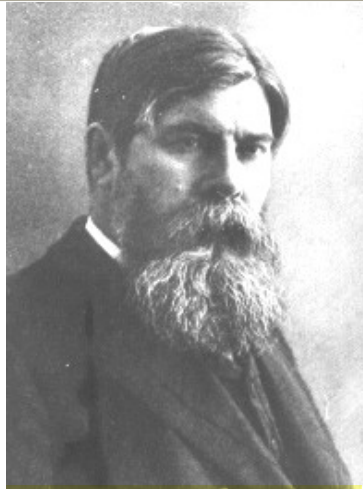
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**RUSSIAN CONTRIBUTIONS****Ivan Pavlov (1849-1936)****Ivan Pavlov**

Ivan Pavlov is one of the greatest psychologists of the 20<sup>th</sup> century whose ideas left a long lasting impact on the field. The importance of Pavlov's discoveries can be gauged from the fact that his theories, as they emerged from his experiments, are today applicable to a very large area of psychology, including:

- Animal/comparative psychology
- Child psychology
- Educational psychology
- Organizational psychology
- Industrial psychology
- Sport psychology
- Psychology of advertising
- Clinical/Abnormal psychology
- Psychotherapy

That is the magnanimity of the impact of Pavlov's discoveries on modern psychology.



Vladimir Bekhterev (1857-1927)

### **Vladimir Bekhterev**

Another contributor from Russia whose name deserves to be mentioned is Vladimir Bekhterev. He was born in 1857 and died in 1927. He studied under Wilhelm Wundt, the psychologist who established the first psychological laboratory at Leipzig, Germany.

One of the major contributions of Bekhterev is his experiments on animals and humans using mild electric shocks. In other words, Bekhterev used electric shocks of low voltage which were not harmful, for humans and animals. He then noted the response of the organisms to the shocks. The idea behind this experiment was to trace the effect on learning. Bekhterev noted that shocks are associated with withdrawal reaction. Withdrawal reaction means that humans and animals tend to move away from the source of the stimulus. In other words, the action of the organism is to avoid the stimulus which was felt harmful by the body. It therefore moved away. This response was called Associative reflex by Bekhterev. He stated that associative reflex was learning which was similar to learning by conditioning as given by Pavlov.

Another contribution of Bekhterev is that he gave the idea of physical energy to explain the phenomenon of consciousness. He proposed that consciousness is a form of physical energy. The energy resulted in the person being aware of himself and his surroundings. He further stated that all higher psychological functions such as thinking and generation of ideas are the forms of expression of physical energy which represented the consciousness.

### IMPACT OF PHYSICAL SCIENCES ON PSYCHOLOGY

As the thinking of man went through various phases of development, psychology also underwent developments. By the end of the 19<sup>th</sup> century and the beginning of the 20<sup>th</sup> century, the world and particularly Europe, had witnessed three earth-shaking events in the last two hundred years. The first one was the French revolution which had resulted in abolishment of feudalism and monarchy, setting an example for Europe and the world to follow. This revolution had a great impact on thinking of man. On the other hand the industrial revolution of the 18<sup>th</sup> century was another ground breaking event of the same era. The Industrial revolution in England, Europe, America and Japan had replaced much of manual labor with machines. New towns were established. The third event was the Soviet Revolution in Russia. In Russia the Bolshevik party had overthrown the monarchy, ended feudalism and abolished the concept of private property. All means of production were nationalized and owned by the state meaning that Socialism was established in the state.

The impact of these three events was that the industrialization spread giving rise to an attitude of encouragement of new ideas, initiative and stress on research. Keeping this brief background in view, let us see what impact some physical sciences had on psychology by looking at the contributions of two scientists.

#### Charles Darwin

One of the most important scientists of the 19<sup>th</sup> century was Charles Darwin. He was born in 1809 in England. He got education there and undertook research in Botany, Zoology or Biology. Therefore, Darwin was basically a man of physical sciences rather than psychology but as we shall see, Darwin related his theories of physical sciences with psychology. Darwin died in 1882.

Charles Darwin traveled and conducted experiments in many parts of the world. He collected samples of plants and observed animals. He discovered that animals in one part of the world were different from other parts of the world, but in areas where the conditions were similar, the animals were also similar. He noted this particularly in tortoise. Based upon his extensive research and observation he proposed that man can be studied the same way as we can study animals.

One of the greatest contributions of Charles Darwin is that he put forward the view that life and survival in the world requires constant struggle and battle against hostile forces. He stated that only those organisms survive who can fight the battle and win while weak organisms perish. This is the law of the nature. This law is called “Survival of the Fittest” which was given by Darwin in an article written by him. Relating it to psychology, Darwin said that psychology is a study of tactics of organism for survival and emotions are an important factor for survival of the organisms. This means that organisms use their emotions to accomplish the task of fighting against the hostile environment and forces.

For example, when an animal shows its teeth to a predator in anger, the predator is scared. The animal is actually expressing its emotions. The emotions may merely to scare the enemy. On the other hand, when an animal screams or cries, it is actually calling others for help. Therefore, according to Darwin, this is how organisms use emotions to survive in the battle of nature. Darwin was able to relate the battle for survival in the nature with the psychology of the organism. This reflects his aim of discovering the link between physiology and psychology.

Another contribution of Charles Darwin is he put forward the view that environment plays a dominant role in determining the psychology of a person. This means that the environment, in which a person lives, influences his thinking, his imagination, the way he behaves and his overall personality. This proposition of Darwin gave rise to the nature versus nurture controversy; which means that it is quite uncertain that whether it is the nature that shapes the thinking and behavior of a human being or it is the bringing up that shapes it. Therefore, man's thinking is either a product of the environment that he lives in, or the training that is imparted onto him. It was later proved that there was a lot of truth in what Darwin proclaimed.

**Gustav Fechner**

Another scientist belonging to the world of physical sciences whose thinking had an impact on the development of psychology was Gustav Fechner. He was a German physicist, who was born in 1801 and died in 1887.

Fechner is known for devising ways of measuring psychological experiences in physical terms, called psycho-physics methods. Psycho-physics methods are methods in which Fechner proposed to measure what a person was experiencing psychologically. The measurement was in physical terms using laws of physics. This was one of the most important contributions of Fechner, apart from other noteworthy contributions.

In order to explain the relationship between psychological experiences and their physical measurement, Fechner proposed the concept of “Sensory Threshold”. According to Fechner, sensory threshold is the minimum stimulation required by a sense organ to register a stimulus. This means that, to which intensity of the stimulus the sensory organs of body would be able to acknowledge the existence of that stimulus. For example at which level of volume, a human ear would be able to detect sound.

Fechner went on to identify the “Absolute threshold” which is a stimulus value that evokes a response 50% of the times. This means that when a stimulus is given repeatedly for 100 times, if it evokes a response 50 times, that is the absolute threshold.

Fechner also identified the “Differential threshold” which is the stimulus value where one stimulus is regarded as different from another stimulus 50% of the time. This means that when two similar stimuli are given repeatedly 100 times, the difference is noted 50 times. This is the differential threshold. Fechner also devised the method of measuring the absolute threshold; it is called the method of minimal changes.

Because of these contributions, particularly because of his inventions of methods of measuring psychological experience in physical terms he has been called the father of psychophysics.

## STRUCTURALIST SCHOOL OF PSYCHOLOGY

Before the arrival of the Structuralists, psychology was already being considered as the study of consciousness. This meant that the entire scope of psychology encompasses the study of man's consciousness. The distinguishing feature of the Structuralists was that they added further to the definition of psychology being the study of consciousness; psychology to them was concerned with the study of the structure of consciousness. Therefore, Structuralists are philosophers/psychologist who defined psychology as the study of consciousness. Further, Structuralists also tried to solve the enigma of a definition of consciousness. It was imperative to give a definition of consciousness in order to substantiate the view point of psychology being the study of structure of consciousness. Therefore, Structuralists put forward the view that consciousness has a definite structure and it can be defined in terms of its structure. The concept of definite structure was given by this school named Structuralists.

The structuralist school has many followers, but two of the very important names who belong to the Structuralist school of thought are Wilhelm Wundt and Edward Bradford Titchner.

### Wilhelm Wundt

Wilhelm Wundt was born in 1832 and died in 1920. Thinking of man evolved with the passage of time as his surroundings change. Earth shaking events like the industrial revolution had caused man to shift towards empiricism. As a result of the work of Russians, Sechenov and Pavlov on discovering laws of learning by conditioning, through experimentation; the contributions of Darwin discovering the relationship between psychology and physical sciences; and Fechner discovering physical ways to measure psychological experiences, psychology was defined as the study of consciousness. Psychology had finally ceased to be a merely speculative science and was gradually becoming an experimental science based on facts and logical conclusions which were drawn through carefully conducted experiments and profound observations.

But most of the credit goes to Wilhelm Wundt who recognized that psychologists must adopt a scientific outlook and adopt experimentation as the methodology, if psychology is to expand and flourish. As mentioned earlier, by this time, events like the French revolution and the Industrial revolution had changed the outlook of man. Man had become more logical in his approach and tried to find reasons for everything. Therefore, based on these grounds, Wundt based his proposition that since speculation could not be the mode of scientific investigation any more; therefore, psychology also has to adopt the new mode of investigation if it has to become a science.

Wilhelm Wundt is known for establishing the first psychology laboratory at Leipzig, Germany, in the year 1875. This was one of the greatest leaps in the transition of speculative outlook to the scientific outlook, facilitated by Wundt.

Wundt defined psychology as the study of consciousness. He further argued, that if consciousness exist, which actually does, then it must have a structure. To elaborate his viewpoint further, he went on to explain the structure of consciousness. According to him, consciousness could be studied by three things:

- **Experimentation:**

Experimentation was the most important mode of scientific investigation which according to Wundt was the need of hour for psychology to develop. Psychologists therefore, needed to conduct experiments which revealed the true nature of consciousness. In this regard, Pavlov was known for his experiments that he conducted on dogs to discover the process of learning.

- **Introspection:**

Although Wundt believed that psychologists should adopt a scientific outlook, he did not reject introspection as a method of gaining knowledge, because he regarded introspection as a mean of looking into and understanding the structure of consciousness.

- Looking at the past history of man that has shaped his consciousness:

Another method of studying the structure of consciousness is by looking at the past history of man. Darwin was the first one to give rise to the nature versus nurture controversy. According to him, it was the environment that shaped the thinking of man, or the way he is brought up. Therefore, according to Wundt, looking into the past of individuals gave insight into the consciousness of the individuals.

Wundt himself also conducted some experiments and based upon them he proposed that consciousness is a process which has three compartments:

- Ideas
- Feelings
- Emotions

He went on to say that the compartment of feelings can be divided as follows:

- Pleasant/unpleasant
- Relaxed/tense
- Calm/excited

Wundt also outlined the abnormal states of consciousness. He then went on to elaborate hallucinations, hypnotic states and dreams. According to him, all of these abnormal activities of the consciousness are caused due to the break down of attention.

Because of his elaboration of various parts of consciousness, he is classified as a structuralist today. He is regarded as one of the greatest psychologists of the year and holds a high position in the history of psychology.

### **Edward Bradford Titchner**

Edward Bradford Titchner was born in 1867 and died in 1927. He was an English man and pupil of Wundt. He studied in Germany, and worked all his life in America.

Contributions of Titchner to psychology can be gauged from three things that he tried to do, are:

- Contents of consciousness

This means that Titchner explained what consciousness is actually composed of. He then went on to elaborate the contents of consciousness.

- Combination of these contents

The second contribution of Titchner is that he described the combination of contents of consciousness which means that, which contents get together to result in an activity.

- Connections between the contents

The third contribution of Titchner is that he explained the connection between the contents of consciousness which means that consciousness is a product of contents being related with each other and working together.

Unlike Wundt, who gave three methods to study consciousness, the subject matter of psychology, Titchner argued that since the only way psychology could be studied was introspection which he defined as systematic and controlled self observation.

This is where he carried forward the view of his teacher, although he did not emphatically propose experimentation as the way of understanding consciousness. Titchner went on to say that introspection is a

special method of gaining knowledge and understanding consciousness and not everyone could introspect. This made him train subjects to introspect.

Wilhelm Wundt, who was a teacher of Titchner, tried to establish a scientific outlook of psychology. But there was a flaw in Titchner's ideas, which was that he took psychology back to the realm of speculation and moved away from the modern empirical approach.

Wundt is also regarded as a Structuralist because he also emphasized on psychology being the study of consciousness and explained the structure of consciousness and described the content of consciousness.

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## FUNCTIONALISM

Various schools of thoughts have emerged in psychology in the last two to three hundred years. These schools differed from one another on the basis of what they reckoned as the subject matter of psychology and how they explained it. One of the early major schools in the development of psychology is the functionalist school.

Like the Structuralist school, Functionalists also defined psychology as the study of consciousness but instead of explaining the structure of consciousness and describing its content, Functionalists focused on, as the name implies, the functioning of consciousness. There are two branches of functionalists:

- o American functionalists; those who were on the American continent.
- o European functionalists; those who were on the European continent.

### American Functionalists

American functionalists are philosophers/psychologists who practiced on the American continent and considered psychology as the study of consciousness, focusing on functions of consciousness.

### William James

William James was born in 1842 and he died in 1910. He was an American philosopher, brother of the novelist Henry James. In 1872 he joined the Harvard faculty as lecturer of anatomy and physiology, after 1880 worked in the department of psychology and philosophy, continued teaching until 1907. In 1890 he published his brilliant and epoch-making book *Principles of Psychology*, in which the seeds of his philosophy are already discernible. James's fascinating style and his broad culture and cosmopolitan outlook made him the most influential American thinker of his era.

James is considered the founder of the sort of thinking called functionalism. The basic question that was raised by James when he purported his theory was that “what is the purpose of consciousness?” Therefore unlike the Structuralists who emphasized on explaining the structure and the parts of consciousness, James came up with a completely new point of view. He focused on why we need consciousness. This need is the function that the consciousness performs, therefore his theories focused on explanation of the functions of consciousness and he is known as the founder of functionalist school.

William James put forward the view that consciousness is not epiphenomena, means that consciousness cannot be considered as something whose functions or working cannot be explained. Consciousness has to have certain set patterns of functions and it shall always work according to those patterns. Therefore, if we consider consciousness to be an “epiphenomena” it would be wrong. Consciousness needs to be studied and understood in order to explain its functions.

Explaining the function of consciousness, James put forward the view that consciousness creates adjustment between the organism and the environment. This means that the organism needs to adjust and adapt its environment in order to survive. This adaptation process is carried out with the help of consciousness. Consciousness helps the organism to understand the environment and cope with the changes in it.

James was also of the view that not just consciousness but all human psychological functions have a definite purpose. For example, fear in an animal creates movement in the body; laughter reduces tension etc. Therefore, the importance of all these psychological functions as well as consciousness is due to their functions. If these functions are disturbed, the organism cannot survive. Therefore, instead of focusing on the structure of the consciousness, scientists should focus on the functions.

His view was based upon his philosophy of pragmatism which means that the validity of an idea is tested by its consequences. In other words, when we need to assay how valid an idea is we need to look at the consequences of the application of idea. If the consequences are as expected, the idea is correct, if not, the idea is wrong. Adopting a pragmatic approach, James was able to give the idea of the functions of consciousness being more important.

William James is also known for his theory of emotions, now called James-Lange theory of emotions. According to this theory, emotions are the consequences of perceptions of bodily changes. For example, a person feels angry because he strikes something. The perception that is generated by this event results in generation of certain responses which are emotions. In this case the emotion would be anger.

He also emphasized that repetition is most important for learning and it creates habits. Therefore, we can develop habits by repeating certain acts. In other words, if we want to develop a habit of something, we need to perform the thing again and again or repeatedly. Ultimately it would become a habit. These were some of the contributions of William James.

### John Dewey

The other prominent American functionalist was a philosopher, psychologist and above all an educationist, John Dewey, born in 1859 and died in 1952. He taught at the universities of Minnesota, Michigan and Chicago and at Columbia from 1904 until his retirement in 1930.

Since John Dewey was an educationist, his theories focused on the field of education. Dewey put forward the view that education, particularly of children, should be based upon the needs of the children. This means that children of different ages have different needs. The education system should concentrate on understanding those needs and should adjust itself to meet the needs. In other words, while forming an education system, the needs have to be focused upon. For further understanding, children have different educational needs during his or her developmental stages. These needs need to be focused upon. This was one of the greatest contributions of John Dewey. His point of view had a great impact on educational practices in the States and globally.

## EUROPEAN FUNCTIONALISTS

### David Katz

The first of the European functionalists was David Katz. He taught psychology at Stockholm, Sweden, for a number of years. In order to develop theory of functions of consciousness, Katz experimented upon how we see colors. He showed through his experiments that the perception of colors remains constant in different lights. This means that if someone is asked about a color in one kind of light, his or her answer would be the same in another kind of light. But in reality, colors change because of different lights. David Katz showed that human subjects see them as before. This he called the “phenomenon of color constancy.” Katz purported that color constancy is a function of the consciousness which makes the person perceive a color in the same way as before. Therefore, Katz also focused on explaining the functions of consciousness and hence included in the functionalist school of thought.

### Edgar Rubin

The other European functionalist was Edgar Rubin who taught at Copenhagen, Holland. Rubin showed by his experiments on perception, that human beings see shapes in terms of figure and ground. This is not just limited to the sense of vision but also on other senses. For example, in case of the sense of vision, sky is the ground and clouds are the figures. When we look at the clouds, we acknowledge their existence because they exist in a background of the blue sky. Further, when we look at a tree, the sky again serves as the background while the tree is the figure in the background. In case of other senses, the sense of hearing also depicts this phenomenon. When we listen to a song, the music being played is the ground and voice of the singer is the figure. Therefore, Edgar Rubin also focused on how the consciousness functions rather than the structure of the consciousness.

### Jean Piaget

The third European functionalist was Jean Piaget who worked at Geneva and Zurich, Switzerland. Piaget is famous for his theory of how children’s minds develop.

He postulated the concept of Schemata. According to him, Schemata is a construct or an idea in the mind of children that how the world is, and how the world functions. In other words, it is what the children perceive about the world around them. They think about how everything occurs, how things go about and how they function. Of all this they make up explanations which according to Piaget is a Schemata.

Schemata develop over the time by two processes:

- Assimilation
- Accommodation

Here, assimilation is the absorption of new and different information about the world and accommodation according to Piaget is the expanding, stretching and changing of schemata to absorb new information about the world. This means that when a child is exposed to various happening of the world he absorbs information from the happenings. This information, collection or absorption process is called assimilation. On the other hand, the information is a subject to constant change. As the child grows, he is exposed to more and more information which requires constant refurbishing of the previously gathered information. This process is called accommodation.

Another contribution of Jean Piaget is that he elaborated how children develop. According to him development takes place in four stages:

- The sensory-motor stage

This is the first stage of development when children learn using their sensations. For example a child touches a hot object and is hurt, this would create awareness in his or her mind that touching hot objects is harmful. Therefore, the child learns by using the sensations. Further all sensations like hot, cold, rough, smooth shall result in learning.

- The pre-operational stage

This is the second stage when the child for the first time, discovers rules and principles of how things work. In other words, the child has the primary or initial information about the working of the things that he sees around him. It is above from learning just through senses.

- The concrete operational stage

This is the third stage when his reason and logic develops and he learns by reasoning. This stage combines the information gathered from the previous two stages and the child is able to develop his own explanation of the things.

- The formal operational stage

This is the most developed and the last stage of development where the child is able to learn by abstraction. In other words, the child is able to draw conclusions about working of things, and phenomena using his own ability to associate ideas, perceive, think, and explain the happenings.

Jean Piaget therefore studied psychological functions in terms of learning and in terms of development. This means that he focused on how learning and development takes place. In other words, he also focused on the functions rather than structure of psychological phenomena. He is therefore included in the functionalist school as a European functionalist

## BEHAVIORISM

One of the greatest developments in the history of mankind is the industrial revolution of the 18<sup>th</sup> century. It marked the advent of a completely new era of thinking and approach towards gaining knowledge. A rational approach was adopted and the ancient speculative methods given by the Greeks were rejected. The industrial revolution had a deep impact on the development of human thoughts; therefore the discipline of psychology was also bound to be impacted.

The path of this influence on psychology is easily traceable. As industry developed and manual labor was shifted to machine labor, there were a number of changes in society and human behavior. New towns were established, close to the factories; housing shifted from huge farm houses to small houses; labor working hours and habits changed; dependence on modes of living changed as well. One of the impacts of the industrial revolution was that, as the production shifted from manual labor to factories, the production increased rapidly. Therefore, new markets were sought and it became imperative to predict and control human behavior, in order to sell more. Another aspect of this development was that the attention of researchers now shifted towards these workers aiming at getting maximum productivity from the industrial labor. This productivity was based on human behavior and the result was that, the need to study human behavior arose. As mentioned earlier, man had become more rational in his approach and had disposed of speculation as a means of gaining knowledge. This made him focus on concrete facts and adopt scientific approach in his study. Psychologists started analyzing, predicting and controlling human behavior since it was visible while consciousness was not. It gave birth to the school of thought now called the Behaviorist School.

### Edward Lee Thorndike

Edward Lee Thorndike was an American behaviorist philosopher/psychologist, who was born in 1874 and died in 1949. After completing his studies Thorndike moved to Harvard University, where William James had set up his psychological laboratory. Thorndike had read James and was impressed with his work on functions of consciousness. He set up his lab at Harvard, working with chicks but later transferred to James' house where he did his experiments on cats which are his most famous experiments. He devised the "puzzle box" which he used for these experiments.

Based upon his experiments he formulated what he called "laws of learning," or how learning takes place. In other words, Thorndike made efforts to learn how the process of learning actually takes place. Before we look at those laws, first let us have a look at the concept of learning that Thorndike proposed. He said that learning takes place by stimulus-response connections rather than by association; according to him learning takes place by two ways which he called laws of learning. The laws are as follows:

#### **i. Law of effect**

The law of effect states that all responses that followed by satisfaction are stamped into an individual and he learns those responses. This means that any act that is performed by an individual, when it is reinforced by a reward that brings satisfaction to the performer, the act becomes learnt.

For example, if an animal such as a cat is rewarded with food if it performs a certain act, the act shall be learnt by the cat. This is what Thorndike concluded through his experiments.

#### **ii. Law of exercise**

The law of exercise states that responses that are repeated are also stamped in and become learnt. This law does not focus on the reinforcement through satisfaction but states that when the individual keeps on performing a certain act, the act is learnt. In other words, the exercise that is constantly performed by an individual is learnt by him or her.

The law of effect given by Thorndike is similar to Pavlov's law of conditioning by reinforcement and the second law, i.e. law of exercise is similar to James's theory of habit formation due to repetition.

As mentioned earlier, Thorndike devised the puzzle box for his experiments. The puzzle box was a box with a door operated through a lever in it. A string with a ring was attached with the lever, which released the door of the box, when pulled. The box was big enough to accommodate a cat. Thorndike used cats for his experiments. As the cat was trapped in the box, it tried hard to escape. The only escape possible was through the door which opened by pulling the string attached to the lever. As the cat pulled the string with its paw, the door of the box opened and it escaped. When the cat was put into the box a number of times, it learnt to use the string to escape. Therefore, based on this observation, Thorndike came up with his laws of learning.

Thorndike also emphasized on a concept important to learning, which he called the "Recency effect." He saw in his experiments on cats that the last act in the series of acts by the cats, which is reinforced, is learnt quickly. For example if a cat is rewarded for the last act which it performs in a series of acts, the last one shall be learnt by the cat. Therefore, as the name implies, the law relates to the most recent act performed by the subject which is learnt by it.

Thorndike performed further experiments to substantiate his laws of learning. This further experimentation and observation lead him to repeal the law of exercise, and modify the law of effect. He then proposed the "law of belongingness" to explain animal learning.

### **iii. Law of belongingness**

The law of belongingness states that only relevant responses that "belong" to the learning situation are learnt. For example, the moving of paws by the cat in the puzzle box is learnt but not pricking of ears.

Thorndike could be classified as a behaviorist because he explained learning as a psychological function in behavioristic, observational terms. He did try to look into consciousness but restricted his studies to the behavior of animals and used comparative psychology to explain human behavior and learning.

## BEHAVIORISM

### Albert Weiss

Albert Weiss was an early behaviorist who was born in 1879 and died in 1931. He was born in Germany but he came to America when he was very young. He studied and worked for the rest of his life in America, therefore he is considered to be an American behaviorist.

The era of the behaviorists was the time when man had adopted a rational approach towards gaining of knowledge and speculation was no more considered as a means for it. This was the time after the industrial revolution which changed the outlook of mankind completely. Writing on psychology, Weiss asserted that anything that cannot be studied with the methods of natural sciences should not be a part of psychology. In other words, anything that is based on mere speculation and cannot be substantiated by concrete evidences should not be a part of psychological study. In other words psychology should concern itself only with observable concrete facts and not with elements, structure or functions of consciousness because these variables are not directly observable. As the Functionalists and Structuralists before the behaviorists had been trying to develop theories about the structure and functions of consciousness, Weiss rejected their approach. To him their methods of explaining phenomena were based on speculative studies.

Weiss further stated that psychologists only pretend when they regard introspection as a method of scientific data collection. This was the method adopted by the schools before the Behaviorists.

One of the important arguments of Weiss is that there are no mental forces apart from physiological forces which are reducible to physical forces. In other words, all mental forces can be studied using a physiological approach and not considering them to be extraordinary forces which cannot be studied through ordinary scientific methods of inquiry.

Another contribution of Albert Weiss is that he also noted the impact of social forces on a person. He said that every person's behavior is impacted by social forces. This means that the social environment, in which an individual lives, influences his or her behavior and thinking. Therefore, he regarded human beings as a product of biological and social factors. He stated that humans are biological and social in nature and psychology is a bio-social science which focuses on relating the biological processes of learning and human behavior with the social environment that is responsible for them.

### Edwin Holt

Edwin Holt was another early behaviorist who was born in 1873 and died in 1946. He was at two of the top universities in America; Harvard and at Princeton. He agreed with the contemporary point of view prevailing on the American continent that psychology should concern itself only with directly observable facts and data. The speculative methods of learning and exploring were rejected by him like all other behaviorists. Therefore, he also declared psychology be a scientific discipline which was based on rationality. Only the facts that could be substantiated should be considered in psychology.

Holt further proposed that psychology should concern itself with behavior, therefore he is also considered as a behaviorist. According to him, behavior is the result of two factors:

#### **i. Learning**

Learning takes place when an organism is exposed to internal or external stimulation, e.g. internal stimulation may be hunger, thirst etc. while external stimulation may be heat, coldness etc. Organism behaves in response to these stimuli.

#### **ii. Canalization**

Further, behavior is also the result of canalization which is what we learn in our childhood. Childhood experiences influence and produce behavior and in any study of psychology we must not ignore those childhood experiences.

### **Walter Hunter**

The third early behaviorist was Walter Hunter, born in 1889 and died in 1953. He proclaimed that he started with dealing in psychology of experience and shifted to psychology of behavior.

He proposed, agreeing with his contemporary psychologists, that psychology should concern itself with observation of behavior of humans and should not concern itself with the problems of consciousness. He said that Wundt was partly correct when he studied consciousness as the subject matter of psychology because consciousness is merely a name used to describe concrete objects in the environment. In other words, a person's consciousness mostly comprises of objects in his environment, which is a view point similar to the one given by Weiss. Environment plays a dominant role in determining a person's behavior and his habits. The prime example relevant here is the impact of the industrial revolution. The example could be related to the behavior of the psychologists themselves. As the environment of mankind changed from speculation to rationality and factualism, psychologists also shifted towards scientific outlook. The need was to have the greatest productivity from the labor force, which was of course related to the study of behavior. Therefore, psychologists shifted towards the study of behavior. The point remains, that the environment and social circumstances have a profound impact on the behavior of the individuals which psychology tends to identify and study.

Hunter chose experimental method as his method of investigation and he is credited with the development of temporal maze for his experiments. Temporal maze was used to conduct experiments in which an animal was allowed to find its way around the maze to the food. When the animal had become familiar with the maze, it would take much less time to reach the food and thus Hunter drew his conclusions based on these observations. He conducted experiments on delayed reaction time of animals for which he is known for. He is also considered as a behaviorist because of his scientific outlook and his emphasis on behavioral approach towards psychology.

## BEHAVIORISM



J.B. Watson (1878-1958)

### J.B. Watson

J.B. Watson is regarded as the founder of the school of behaviorism. He was born in 1878 and died in 1958. He began as a student of philosophy at the University of Chicago, but later turned to psychology. He taught for a number of years at Johns Hopkins University at Baltimore USA, where he set up his animal laboratories. Later he shifted to the corporate world and offered advice to industry relating to advertising and marketing.

Watson taught that psychology should ignore consciousness and concentrate on concrete facts: psychology. This was one of the founding principles of the behaviorists' approach. He further said that psychologists must discard all reference to consciousness and must only look at behavior of animals and man. Because of his stress on behavior to the neglect of consciousness, he called himself a behaviorist. He was the first one to proclaim himself as a behaviorist. His methodology revolutionized the subject of psychology giving a new outlook to it. As consciousness was no more regarded as the concrete method of gaining knowledge, therefore, psychology focused merely on factual evidences and observable phenomena after the advent of this school. A measure of how seriously his appeal was taken by his professional colleagues is that he was elected as the President of American Psychological Association.

In one of his books, entitled "Behavior," he enumerated what behaviorism is all about in psychology. As the functionalists and the structuralists had defined psychology as the study of consciousness, Watson defined, as opposed to them, psychology as the science of behavior. Behavior of animals and humans was in his view what needed to be studied for an understanding of psychology.

Furthermore, Watson asserted that psychologists should use only objective, experimental methods and should not use introspection as a method. He said that the aim of the study of psychology should be to provide prediction and control of behavior. This is the basic aim of behaviorism. Behaviorists tend to develop methods and techniques to control and predict human behavior in order to get the most out of them. Behaviorism emerged in times when the industrial revolution took place. At that moment in the history of mankind, the focus was on increasing the productivity of workforce.

According to Watsonian behaviorism, behavior can be studied in terms of stimulus-response patterns. This means, that a stimulus is received by organism and it responds. For example, when someone touches a hot object, he immediately withdraws his hand from the object. In other words, the hotness of

the object serves as the stimulus while the withdrawing action of the individual is his or her response to the stimulus. Watson therefore stated that there is nothing mysterious in this action and reaction and all of it could be explained in simple physiological terms.

Watson denied the value of introspection as data for psychology but said that a “verbal report,” may be obtained from the subject after the experiment. For example, if an individual is placed in a series of experiments, he may then be asked about the feelings and the emotions that he faced during the experiments. It is different from introspection in the sense that here the report is based on the circumstances that have been artificially created for the experiment. Therefore, verbal report, in view of Watson may be a source of information for psychologists, but he clearly denied the introspection as a means.

One of the important contributions of Watson is that in his opinion, thinking is nothing but “implicit behavior.” For example, when an individual manipulates images in his mind, thinking takes place. The individual relates these images together forming an explanation for the phenomenon that he is trying to study. Therefore thinking is also a kind of behavior. Watson agreed with the viewpoint of Ivan Pavlov about learning and said that we learn according to the laws of conditioning as given by Pavlov.

Watson suggested that memory and images are nothing but sensory activities in the brain. This again refers to the study of behaviorism as a physiological phenomenon rather than a mysterious one. He said that the sensory activities of the brain can be classified as “molecular behavior.”

Watson further proclaimed that by controlling the environment of an organism we could control and predict its behavior. This is known as environmentalism, that organism is affected by its environment. It is similar to the idea of Tabula Rasa given by John Locke. According to Locke, the mind of a new born baby is like a clean slate which is written upon by the surrounding environment. Watson’s contribution to psychology was one of the major developments in the study of psychology.

### NEO-BEHAVIOURISTS

Neo-Behaviourists are psychologists who adopted the line of thinking and methodology of Watson, but were born or worked after Watson. Watsonian behaviourism gave a new direction to the science of psychology. The old method of introspection and speculation as a means of information for psychology was rejected and a more scientific outlook was adopted. Only concrete facts were considered which were observable. Therefore, behaviourism became very much as the heart of psychological thought. Neo-behaviourists followed the Watsonian line of thinking although they were born after Watson.

#### Clark Hull

The first neo-behaviourist was Clark Hull who was born in 1884 and died in 1952. He was an engineer and a mathematician but later turned into a psychologist. He taught at the University of Wisconsin. Hull attended Koffka's lecture on Gestalt psychology at Wisconsin University and was impressed with the Gestalt School. He had also read Pavlov's English translation of "Conditioned Reflexes," and liked it very much. His explanation of human behaviour therefore reflects both the Gestalt and the Pavlovian points of view.

Since Clark Hull was basically a mathematician, he aimed at expressing his views in mathematical terms very precisely. He gave a mathematical equation which explained his view point. The equation is as follows:

$$sE_r = D \times V \times K \times sH_r$$

Where:

sE<sub>r</sub>, is behaviour

D, is drive, our need within

V, is Valence or stimulus intensity

K, is incentive, motivation

sH<sub>r</sub> is habit

Therefore, according to Hull, behaviour is dependent upon drive, valence, incentive and habit. Since Hull was a behaviourist he tried to simplify behaviour and according to him, as a person acts in a certain manner or behaves in a certain manner, the driving forces behind it are D, V, K and sH<sub>r</sub>. Drive according to Hull is the need inside a person. A person may want to achieve a certain goal, reach a target or perhaps want to have some possession for him. All these things constitute the drive inside the person are expressed as D in Hull's equation.

V is the intensity of the stimulus that governs certain behaviour. In other words, it is how strongly the stimulus is registered by the sense organs of the body.

K is the incentive or the motivation. It is the target or the reward that the individual or the subject aims at. It serves as the incentive for the subject to behave in a certain manner which would allow him to reach the reward.

The last component of Hull's equation of behaviour is sH<sub>r</sub>, which is the habit that a person develops. Behaviourists had given the explanations for why habits are developed. Hull carried forward their ideas and said that habits also serve in determining how a person behaves.

In simple terms behaviour is the result of many factors that include D: drive, V: attraction or repulsion, K: motivation and habit. As it can be seen, his theory rests mainly on looking at performance, on observing overall overt behaviour and that is why he is classified as a behaviourist.

### **Edward Tolman**

Edward Tolman was another American, who was born in 1886 and died in 1961. He studied at Harvard and was impressed by William James and Watson. Tolman believed that psychology is a science of behaviour and psychologists should concentrate on the study of behaviour only and in addition to it he said that behaviour has a “purpose.” That is why he is also called a “purposive behaviourist.”

He defined behaviour in a precise equation that is:

$$B = f(S, A)$$

Where:

B is behaviour

f is the function of

S is situation variables

A is antecedent variables

This means that behaviour is a function of situational variables and antecedent variables.

**A situational variable** is any variable that is a part of the situation/environment that affects a subject's behaviour in a way, for example the hot or cold weather.

**An antecedent variable** is a variable that is a part of the subject such as age, gender etc. and it impacts its behaviour. Therefore, his explanation of behaviour was very precise. Because of his theory of explanation of behaviour, Tolman is classified as a neo-behaviourist.

### **Edwin Guthrie**

Edwin Guthrie was also a prominent neo-behaviourist who was born in 1886 and died in 1960.

Unlike his contemporary psychologists, Guthrie was less an experimentalist and more an observer. Based upon his observations, he put forward the view that behaviour can be predicted and controlled by the “law of recency,” given by Thorndike, which states that “the last act is most likely to be repeated.” Therefore, behaviourism to Guthrie is repetition of the “last act.” Thorndike proposed his law of recency based on experiments that he conducted on cats. Therefore, the basis of Guthrie's thoughts resides in experimentalism.

Guthrie had also read Freud and was impressed by his method of treatment of mental disorders. Based upon his behaviourist outlook, Guthrie developed a theory of neurosis in his book “The Psychology of Human Conflict” and because of this he may be regarded as one of the first behaviourist psychotherapists. He explained neurosis to be caused due to conflict of responses.

## NEO-BEHAVIORISTS

### B.F. Skinner

One of the most prominent psychologists, who was impressed by Watsonian behaviourism on the one hand and Pavlovian conditioning on the other was another American B.F. Skinner, born in the year 1904 and died in 1990. He worked at Harvard University and carried on his experiments on animals, writing many books and articles. His main research work is now known by the title of Instrumental or Operant Conditioning.

Working on white rats and pigeons, in specially devised cages, known as Skinner boxes he observed how animals learn. He therefore adopted an experimental method of learning. He also purported the idea of learning by conditioning but his conditioning was different from that of Pavlov. Later Pavlov's conditioning came to be known as classical conditioning while Skinner's conditioning was called instrumental or operant conditioning.

One of the contributions of B.F. Skinner is that he distinguished between positive reinforcement and negative reinforcement, where positive reinforcement is when a response is positively rewarded and negative reinforcement is when a response is negatively rewarded or punished. In other words, if a subject, who is hungry, is rewarded on its act with food, the reinforcement is positive. On the other hand, if a subject is punished on his or her act but being beaten up or being subjected to some kind of torture, or even a less severe punishment such as denial of food, the reinforcement given to him according to Skinner is negative reinforcement.

According to Skinner, learning takes place by four schedules of reinforcement:

- i. Fixed ratio schedule
- ii. Variable ratio schedule
- iii. Fixed interval schedule
- iv. Variable interval schedule

Reinforcement is given to the subjects according to these scales. Fixed ratio is, when reinforcement is given after a fixed number of responses. Variable ratio is when reinforcement is given after variable number of responses. Fixed interval is reinforcement is given after a fixed time period. Variable interval is when reinforcement is given after a variable time interval.

Based upon his experiments he proposed in his famous book: "Beyond Freedom and Dignity" that change in whole society is necessary for the betterment.

### Karl Lashley

An important contributor to the behaviourist school is Karl Lashley, born in 1890 and died in 1958. He was a pupil of Watson. Adopting from Watson, he proposed that psychology is a science of behaviour and behaviour is the result of activity of the cerebral cortex, a part of the brain. This is one of his greatest contributions in psychology which lead to the development of two principles given by Lashley himself.

- i. Principle of equi-potentiality:

According to Lashley, one part of the cortex is the same as another part. Therefore, even if one part is cut off, the brain activity still takes place. In other words, both parts of the cortex have equal potential to make the brain work properly.

**ii. Principle of mass action:**

He also proposed that the more the cortex in mass, the better the learning and called it the “principle of mass action.” So Lashley showed that behaviour and more precisely learning is a function of the activity of the cortex.

**Donald Hebb**

Donald Hebb was a Canadian psychologist, who showed that, when a part of the human brain was removed accidentally, it did not affect a person’s IQ. It is similar to Lashley’s principle of equi-potentiality which stated that if one part of the cortex is removed, the brain continues to function as with the full cortex.

**Harvey Harlow**

Another psychologist Harvey Harlow, of University of Wisconsin showed that curiosity, exploration and manipulation effect animal learning. This meant that if a subject is curious about certain phenomenon, it would have a different learning pattern as compared to other subjects. Further, exploration and manipulation also affect learning.

**Hobart Mowrer**

Hobart Mowrer at the University of Illinois showed that factors such as hope and disappointment influence learning.

Behaviourism slowly had introduced such mentalistic concepts as curiosity, hope, disappointment, etc. So behaviourism slowly moved away from dealing with purely observational data and started looking into psychological factors.

## GESTALT PSYCHOLOGY

There were two major trends in psychology at the end of the nineteenth century and the beginning of the 20th century. In Europe, Wundt's structuralist psychology was the major influence, and in America behaviorist psychology was the dominant trend. In 1912 three German psychologists located in and around Frankfurt, Germany, independently of each other had come to the conclusion that in the past psychologists had overlooked the linkage between consciousness and behavior. Therefore these psychologists started to concert efforts to develop a system of psychology that took into view the unity of human beings. These efforts laid the foundation of a new look called Gestalt psychology, the purpose of which was to avoid piecemeal study of psychology and to present a holistic view. In other words, the Gestalt psychology adopted both the behaviorist's views and the structuralists' views. Gestalt is a German word meaning patterns. The structuralists were focusing on the consciousness as the key element in the study of psychology while the behaviorists were focusing on the prediction and control of behavior. The group of students and researchers related to this concept developed the Gestalt School.

### Max Wertheimer

Max Wertheimer belonged to the Gestalt school of psychology. He was born in 1886 and died in 1943. He studied at the universities of Prague, Berlin. His original researches, while he was a professor at Frankfurt and Berlin, placed him in the forefront of contemporary psychology. Wertheimer came to the United States in 1933, shortly before the Nazis seized power in Germany. Wertheimer's discovery (1910–12) of the phi-phenomenon (concerning the illusion of motion) gave rise to the influential school of Gestalt psychology. His early experiments, in collaboration with Wolfgang Köhler and Kurt Koffka, introduced a new approach (macroscopic as opposed to microscopic) to the study of psychological problems. In the latter part of his life he directed much of his attention to the problem of learning; this research resulted in a book, posthumously published, called *Productive Thinking*.

One of the greatest contributions of Wertheimer is that he showed by his experiments that if two lines are shown to a subject and the time period of exposure between these two lines is small, the subject sees these two lines as one line moving from its position to the position of the other line. Wertheimer called this phenomenon "Apparent movement" or phi-phenomenon. Therefore, according to Wertheimer, Phi-phenomenon or apparent movement is when we see one image move from one place to another, when physically there is no movement. In case of the lines shown by Wertheimer, the horizontal or the vertical lines did not move at all, but instead, one line appeared after the other. The interval between the disappearance and the appearance of the other line was so short that to the subjects it appeared that the lines were moving from their positions into the other positions. The greatest impact of this discovery can be seen in its application in the movie camera. When a movie is being played on a cinema screen it is actually the phi-phenomenon which is working. The images of the movie are all still images captured by the camera but the interval between the movements of the images is so short that the characters of the movie seem as moving to the observer. Television is another example where the image is created by a small dot which moves across the screen and the characters appear to be moving.

Wertheimer explained this phi-phenomenon as being due to a tendency on the part of human mind to fill in the gaps. For further explanation, when the line which was perceived by the observer as moving is analyzed, we see that the movement from the horizontal to the vertical or vice versa, has been developed by the observer's brain, while there is no movement at all. Therefore human mind has the tendency to develop something to fill into the gap. Based upon this tendency, Wertheimer discovered many factors which help human beings to perceive things in patterns or Gestalts. He called these factors, the factors of organization. They are factors that help us to perceive in patterns or Gestalts. Some of these factors are:

### ○ Similarity

If you see many dots and small lines, you see dots as one group, pattern, gestalt etc. and lines as another group. This is similarity leading to gestalts or patterns based on groups.

### ○ Proximity

If you see many dots close to each other, and then some dots separated, you see the closely located dots as one group of gestalt. This is nearness or proximity leading to gestalt.

### ○ Closure

If a part of a familiar shape is missing we tend to fill it up, and see the shape as whole, this is closure leading to gestalt. For example, if one of the corners of a star is missing, we tend to fill in the missing part by ourselves and perceive the star as its complete shape. Therefore closure also results in gestalt.

These are factors that are in the stimulus field that help us to perceive gestalts. There are some subjective factors also that help in this whole perception. For example, mental set, or set is a subjective factor that helps perceptual organization. Mental set of set can be explained with the help of the following examples: if a person is taking a walk in a garden, and before he came here, he was warned by his friend that there were snakes in the garden; his mental set would be to see snakes in the garden. Therefore, he is likely to confuse a twig with a snake and be afraid of it. This is an example of mental set.

Habit or familiarity is another factor that leads to perceptual organization. For example, if a person is familiar with certain objects he may be able to formulate a gestalt very quickly. Same is the case with habit. Wertheimer, through his observation and experimentation discovered those factors that influence perception.

Wertheimer also tried to discover what is creative thinking or problem solving thinking. Creative thinking or problem solving thinking had become a subject of interest for psychologists at the turn of the century since creative thinking was the key to development as newer fields of study were explored by mankind. He observed young children and adults in his quest to determine what is creative or problem solving thinking and how it takes place. He also interviewed one of the greatest minds of the 20th century, Albert Einstein, to see how he produced his Theory of Relativity. Based upon these observations, Wertheimer noted various operations related to creative or problem solving thinking. He said we should avoid a piecemeal approach, not let our biases affect our thinking and should not blindly follow our habits. In other words, he said that we should ensure that our dispositions do not affect our thinking and we are able to concentrate on discovering new rather than analyzing new from the already existing point of views. That is how we can become productive, creative thinkers.

Wertheimer is known as a **Gestalt psychologist** because he tried to determine the patterns of perception that an individual follows.

REMEMBER IN PRAYERS  
KEEP SMILING  
SAY NO TO CURRUPTION  
HELP TO OTHERS

Regards,  
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